Command of Evidence — Part 3 answer key

#	ID	Answer	Difficulty
3.1	403a3013-0690-457d-bd78-5aba11ff4960	D	Hard
3.2	f42209ba-02ff-43a9-a852-dfd9f7ce56da	A	Hard
3.3	62e9e4db-75f4-462a-a8d8-577faa5c6eb0	A	Hard
3.4	a69e4082-0554-4a04-b36b-7fa6f918ebbc	B	Hard
3.5	fd1571d8-4dd2-4b3e-acc1-33b0974356d2	D	Hard
3.6	b500714d-33c9-4d4b-81c0-9204cb352618	D	Hard
3.7	f6866682-b170-4251-9fd8-8ce1ae56407d	B	Hard
3.8	de524c7e-6530-4cc4-889d-5f33fb362adc	C	Hard
3.9	346facbf-2c87-4295-94bd-099a869cb95a	C	Hard
3.10	7d889d5e-6ff0-4c2b-a1b2-9b0d56817fea	B	Hard
3.11	c3d04fab-85b3-4607-9920-c2490bdcc643	C	Hard
3.12	483cec0f-fa36-49b0-b170-d9f66685471a	B	Hard
3.13	af31476c-fbc2-4fe8-8656-e63f24aafb95	B	Hard
3.14	a46a3369-477e-4477-972e-bc12eb5e84d2	B	Hard
3.15	cb36de75-9ada-404a-b492-e714308354bb	D	Hard
3.16	c776357f-bef8-451f-ad4f-fcec70402ee7	D	Hard
3.17	b5f97482-9020-4177-bb87-125db8448121	A	Hard
3.18	59ef7528-4cfb-4862-9d40-6f226b28ad3f	A	Hard
3.19	0c85a491-b942-44d2-9407-d63cefec2ad9	B	Hard
3.20	a834b9d5-442c-4133-b51b-b1e6ce7c584c	B	Hard
3.21	76c41880-b3cd-4f49-86f6-2e9fb2b14aa9	B	Hard
3.22	93d69a24-a340-41cd-a1f4-68f2d6b98e24	C	Hard

3.23	f88c0e77-479a-4c8d-996f-c9b7d9e730d3	B	Hard
3.24	05b51061-19d4-45fe-8086-a6b565cc8902	D	Hard
3.25	b31ad613-241a-46ec-a93e-8512bfcf837d	D	Hard
3.26	69ccacae-98b0-4823-a0df-dbc2f84a63fa	C	Hard
3.27	691aa9c9-6158-43be-97a2-6230c41ccfe3	C	Hard
3.28	9d1ab342-87cd-4c5f-8c7e-dbd44981b5e0	D	Hard
3.29	68ed81e3-74c6-4b34-9287-70290ef8046f	A	Hard
3.30	e106d6c6-3728-41aa-983d-844c3f37d99e	D	Hard
3.31	71c4f33b-5e73-413d-bea3-14225980d694	D	Hard
3.32	ce4f3e39-cfd4-4f81-8349-6bbebc6ad702	B	Hard
3.33	ecf8fb3a-a8a8-4251-9424-048c185cf125	C	Hard
3.34	a9859e96-a39d-4207-8cdb-6fb0ee8d8a98	B	Hard
3.35	126dccc3-ece2-4b28-9d66-0e6d7f22c69f	B	Hard
3.36	084d8f9f-c1fe-4c1b-8c34-e8d1f0cb8182	B	Hard
3.37	6fec40ff-b4f1-4439-8f08-ba70ddaff0ed	B	Hard
3.38	d9511af2-0fb9-462e-977e-9833b95dc982	C	Hard
3.39	41e246d6-ff3c-448d-8852-1127eaf889c0	A	Hard
3.40	217d6208-9618-4749-b9af-b89fbf69da88	D	Hard
3.41	0e07e7fd-58a8-43aa-8948-44628be1e76e	B	Hard
3.42	3999df4f-831b-45be-a3b3-2fa658b23a70	B	Hard
3.43	54eed2e9-47bc-4ec4-a3ea-81d4a8cf0a81	D	Hard
3.44	fd9cc9d2-2e39-495a-af16-5f06049dbbc2	C	Hard
3.45	15287480-0c69-412a-a5ec-856eedeac05e	A	Hard
3.46	d1d0c33d-ec34-4d8a-b2d4-4b8262f22cbc	D	Hard
3.47	11ca9379-821f-4d74-ac38-8830e1fc7d08	D	Hard
3.48	704122ea-86c8-4130-a534-dde9ec1a52b1	D	Hard

3.49	675fc7a8-a3ec-4cc1-a704-a49f4644cf26	B	Hard
3.50	f0455dd3-fa60-4155-8945-bd6f8ee836ef	C	Hard
3.51	74e5885a-cf6e-4b0a-b09a-04d8579c9935	B	Hard
3.52	5b18343d-bb14-442d-a3c3-8d74851bbddb	D	Hard
3.53	c7238400-e5ab-4ee8-a20c-29043d59a1a1	C	Hard
3.54	03bcea8c-439e-455a-bb85-7a05d2ed8773	A	Hard
3.55	74ac6d75-993e-42c6-95ac-056e55a93a5e	A	Hard
3.56	65d3f00d-f48c-4518-88b0-b5d58cbe4cf9	C	Hard
3.57	b618a704-36f0-4ec0-a44e-f8d2a7f54307	A	Hard
3.58	40e7dbe5-a37f-471d-bf2a-66c7e950d5d3	B	Hard
3.59	9ecc06d7-b381-4b61-8477-2a34b8437476	A	Hard
3.60	738288ff-e395-4313-835d-9e9447a865b5	A	Hard
3.61	be347d28-4be3-4fe4-b8ec-28dab78f77a8	C	Hard
3.62	e53f5690-706d-4cca-b2f6-dbd34999ecbc	D	Hard
3.63	a9fcb8ca-cf0c-4d45-bcc6-e3fd7a4ee3dd	D	Hard
3.64	1da0f5f5-e5b6-40f5-b42f-81390b87a253	D	Hard
3.65	de7f8b22-30e6-43b1-ad24-4b6f6f7e025f	A	Hard
3.66	7ba4dc16-ed92-4253-b979-eafdfe5400f2	D	Hard
3.67	20bd7e10-d0b5-4923-930e-f3c22dff2c07	D	Hard
3.68	3d1f1c40-4ed9-4ae6-9d5c-55bbe74e0b51	D	Hard
3.69	53727c14-2e6d-4c01-a008-3dcf1597f23d	B	Hard
3.70	a48b7a71-d74f-4589-9010-e1a7a3afac2d	C	Hard
3.71	66352cf4-a0a7-4c66-8a46-9348e6f8d2b5	C	Hard
3.72	23b0f654-fb5e-48f3-8739-811562c9bf99	D	Hard
3.73	0f42ebb1-ab1a-4389-9ce6-727db51e5379	B	Hard
3.74	de355f6b-26f2-4286-9e24-fe61e14d1a86	A	Hard

3.75	1952d001-606f-4761-96e6-f22a2bd7b6c1	D	Hard
3.76	a197aae6-b0b0-47ba-910d-2407eaacd03a	C	Hard
3.77	9a5b800c-9469-4023-8098-70ef352fd380	B	Hard
3.78	48df5d1e-9a25-416a-a935-574134ee86cb	B	Hard
3.79	c4cfbca4-fcc1-4015-a43f-bc8038fe1c6c	A	Hard
3.80	68c34ac6-b03e-440f-82e7-fa34d16e48a9	D	Hard
3.81	870865eb-d11e-4e29-8d5e-e7d9f282da86	D	Hard
3.82	89896d3d-375d-4ea9-ac2a-c00eb083d581	D	Hard
3.83	2a2fa5ca-6a3d-478f-8ee6-9d4532b76393	A	Hard
3.84	7a847714-4f4b-4c3d-b19d-820a04e30eeb	B	Hard
3.85	3275ff87-da2f-478f-b3a4-1e2626b25e08	B	Hard
3.86	0c6b9256-165f-4981-9503-ecfd319b2731	C	Hard
3.87	cebd824f-b240-47cf-8333-e2cff8bd0761	A	Hard
3.88	c43e221a-8ffb-4239-be7d-be41175df72d	B	Hard
3.89	6a2a163b-95a8-46e0-80b5-f33eeff84503	A	Hard
3.90	24f0aa79-c7cc-4438-84e1-bec4e8d2a77f	D	Hard
3.91	30f294ab-b2a2-4e6f-bfc7-f099c4e68738	A	Hard

Rationales

▼ 3.1 — D

Choice D is the best answer. The claim is that divided Congresses are necessary but insufficient—that is, we need divide Congresses, but they are not enough—to decrease government size, as measured by total federal outlays. This choice accurately expresses the supporting data from the “change in total outlays” part of the graph. Within the data set, divided Congresses sometimes decreased total outlays, but undivided ones never did.

Choice A is incorrect. The claim is only about government size, as measured by total federal outlays—defense and nondefense outlays aren’t relevant. Choice B is incorrect. The claim is

only about government size as measured by total federal outlays—nondefense outlays aren't relevant. Choice C is incorrect. The claim is only about government size as measured by total federal outlays—specific information about defense or nondefense outlays isn't relevant.

▼ 3.2 — A

Choice A is the best answer because it most effectively illustrates the theater historian's claim that El Teatro Campesino's early performances were influenced stylistically by *carpa* theater, vaudeville-style shows with "audience-mediated slapstick comedy." The quotation indicates that El Teatro presented comedic sketches that often featured exaggerated physical humor and that the performers improvised based on the cheers and boos from the audience; thus, it illustrates that El Teatro's performances were similar to *carpa* theater in the physical, slapstick nature of their comedy and in the way the performances were mediated, or shaped, by the audience's reactions.

Choice B is incorrect. Although the quotation refers to El Teatro's use of comedy and mentions audiences, it focuses on the company's use of comedic aspects of reality as a tool to critique society while providing audiences with entertainment and inspiration; it doesn't give any indication that El Teatro's performances featured slapstick, or very physical, comedy or that they were directly shaped by the audience. Choice C is incorrect. Although the quotation refers to El Teatro's aesthetic, or style, and mentions comedy, it indicates only that the company relied on satire and humor and had a style that reflected their limited financial means and sociopolitical message; it doesn't give any indication that El Teatro's performances featured slapstick, or very physical, comedy or that they were directly shaped by the audience. Choice D is incorrect. Although the quotation refers to El Teatro's performance of comedy, it mentions only that the company performed brief humorous vignettes, or short humorous scenes, in farm fields; it doesn't give any indication that El Teatro's performances featured slapstick, or very physical, comedy or that they were directly shaped by the audience.

▼ 3.3 — A

Choice A is the best answer. The table provides evidence that the words for corn in the Cherokee and Muskogean languages appear unrelated to those in Caddo language, which is described as closely related to the Totozoquean language family. This supports the claim that corn cultivation spread across the Southeast without necessarily spreading Totozoquean vocabulary along with it.

Choice B is incorrect. This choice emphasizes the diversity of Southeastern languages in general, but we're specifically looking for information about words associated with corn. Choice C is incorrect. This choice shows how words for corn can change and evolve, but we're looking for an example that shows how some words for corn can be completely unrelated. Choice D is incorrect. This choice shows words for corn that share a common root, but we're looking for an example that shows how some words for corn can be unrelated.

▼ 3.4 — B

Choice B is the best answer because it most effectively illustrates the claim in the text that in describing the teenaged girl, Mansfield contrasts the character's pleasant appearance with her unpleasant attitude. In the quotation, Mansfield describes the teenager as having a "lovely nose" (a compliment about her appearance) but also as treating her makeup puff "as though she loathed it" (a judgment suggesting her unpleasant attitude).

Choice A is incorrect because the teenager´s reaction to the flowers doesn't make it clear that she has an unpleasant attitude, and nothing in the quotation indicates that any part of her appearance is pleasant. Choice C is incorrect because the quotation suggests that the teenager has an unpleasant attitude (being upset with the location and leaving the table before the narrator has paid for the meal) but doesn't give any indication that she has a pleasant appearance. Choice D is incorrect because the quotation suggests that the teenager may have an unpleasant attitude (lowering her eyes, wincing, and sitting in silence) but doesn't give any indication that any part of her appearance is pleasant.

▼ 3.5 — D

Choice D is the best answer. This finding, if true, would support the archaeologists' conclusion. Dense central settlements linked to smaller ones would provide evidence of cities and suburbs—in other words, "a form of urbanism."

Choice A is incorrect. This finding, if true, would weaken the archaeologists' conclusion. Widely separated, small settlements with jungle in between would support the long-held belief that the Casarabe culture "was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness." Choice B is incorrect. This choice wouldn't support the researchers' conclusion. These large cities are located outside Casarabe territory, which doesn't show evidence of Casarabe urbanism. Choice C is incorrect. This finding wouldn't support the archaeologists' conclusion. Scattered small farms

in jungle clearings are not good evidence to support the existence of cities ("a form of urbanism").

▼ 3.6 — D

Choice D is the best answer because it most effectively uses data from the table to complete the statement about the researchers' conclusion about fossil shark teeth. The text states that mammalian fossil evidence has placed both MFS and PFBV within Hh4 (6 Ma to 4.75 Ma) but "a more precise determination of the sites' ages has proved challenging," prompting Killingsworth and colleagues to attempt to date the sites by comparing average $^{87}Sr/^{86}Sr$ ratios for fossil shark teeth from MFS and PFBV to ratios in the global strontium seawater curve. Given that "$^{87}Sr/^{86}Sr$ ratios in seawater correspond to numerical ages," a fossil whose $^{87}Sr/^{86}Sr$ ratio corresponds to the ratio associated with a particular age in the global strontium seawater curve can be inferred to date to that age. According to the table, the $^{87}Sr/^{86}Sr$ ratio of 0.709000 for fossil shark teeth from MFS corresponds to an age of 5.86 Ma, which falls within Hh4. By contrast, the table shows that the $^{87}Sr/^{86}Sr$ ratio of 0.709028 for fossils from PFBV would fall between 0.709020 (5.40 Ma) and 0.709040 (4.75 Ma), indicating an age that falls within Hh4 but is more recent than MFS's age of 5.86 Ma. The data therefore corroborate the conclusion that both sites fall within Hh4 while suggesting that PFBV was likely deposited more recently than MFS was.

Choice A is incorrect because the table indicates that MFS was deposited closer to the Hh3-Hh4 boundary than PFBV was, not the reverse. The table indicates that PFBV's $^{87}Sr/^{86}Sr$ ratio of 0.709028 corresponds to an age between 4.75 Ma and 5.40 Ma, while MFS's $^{87}Sr/^{86}Sr$ ratio of 0.709000 corresponds to an age of 5.86 Ma. Because the Hh3-Hh4 boundary occurs at 6 Ma, MFS was deposited closer to that boundary than PFBV was. In addition, the text doesn't establish that mammalian fossil evidence offers less dating precision than $^{87}Sr/^{86}Sr$ ratios do; it states only that mammalian fossils have placed both MFS and PFBV in Hh4 and that more precise dating has proved challenging. Choice B is incorrect because it mischaracterizes the level of support for the site age estimates established through mammalian fossil evidence. The table indicates that the average $^{87}Sr/^{86}Sr$ ratios from MFS (0.709000) and PFBV (0.709028) both correspond to ages that fall within Hh4 (6 Ma to 4.75 Ma)—5.86 Ma for MFS and an age between 5.40 Ma and 4.75 Ma for PFBV. The $^{87}Sr/^{86}Sr$ ratios therefore fully support, rather than only partially support, the previous mammalian fossil evidence placing both sites within Hh4. Choice C is incorrect because the table indicates that the average $^{87}Sr/^{86}Sr$ ratios in fossil shark teeth from MFS (0.709000) and PFBV (0.709028) correspond to different ages—5.86 Ma for MFS and an age between 5.40 Ma and 4.75 Ma for PFBV. Because the ratios point to different ages, they don't support the

conclusion that the fossils at the two sites were deposited contemporaneously, or at the same time.

▼ 3.7 — B

Choice B is the best answer because it presents a finding that, if true, would most directly support Dumas Gálvez and colleagues' conclusion that the venom produced by the scorpion species *Centruroides granosus* helps mitigate, or reduce, risk during feeding in addition to enhancing predation success. According to the text, Gálvez and colleagues found that the venom used by *C. granosus* both subdues prey and inhibits the growth of *Escherichia coli*, a pathogen that, if ingested by *C. granosus*, would presumably cause disease unless neutralized in some way. If it were true that *E. coli* is commonly found in species preyed on by *C. granosus* and, moreover, can withstand *C. granosus*'s digestive juices after ingestion, this would suggest that *C. granosus* likely relies on another mechanism to neutralize *E. coli* to make *E. coli*-infected prey safe for consumption by the scorpion species. Given that, as the text states, *C. granosus*'s venom was found to inhibit the pathogen's growth, it therefore follows that the venom provides protection for *C. granosus* against *E. coli* that its digestive system wouldn't otherwise provide, making it reasonable to conclude that the trait may have spread through the species because it mitigates risk during feeding.

Choice A is incorrect because a finding that *E. coli* doesn't appear to be virulent, or dangerous, for *C. granosus* even when venom isn't used would weaken rather than strengthen Gálvez and colleagues' conclusion that a particular form of venom production spread in *C. granosus* in part because it helps reduce risk during feeing, since this finding would suggest that *E. coli* isn't actually a risk to *C. granosus* when consuming prey. Choice C is incorrect because a finding that *C. granosus* can detect and avoid consuming *E. coli*-infected prey in the first place would suggest that an ability other than venom production is the primary factor that reduces *C. granosus*'s risk when feeding, which would suggest there has been less evolutionary pressure to develop venom that provides protection; thus, this finding wouldn't clearly support the conclusion that *C. granosus*'s form of venom production spread in the species in part because it helps reduce risk during feeding. Choice D is incorrect because Gálvez and colleagues' conclusion focuses on *C. granosus* venom in relation to the risk of *E. coli*, and a finding that the venom also inhibits nonpathogenic (not disease-causing) bacteria species, which presumably don't pose a risk if consumed, wouldn't be relevant to the conclusion that *C. granosus*'s particular form of venom production spread in the species in part because it helps reduce risk during feeding.

▼ 3.8 — C

Choice C is the best answer because it presents a finding that, if true, would weaken the astronomers' claim about the makeup of host stars and their planets. The text explains that because stars and planets begin forming from the same gas and dust, astronomers believe planets should be composed of the same materials as their host stars, but in equal or smaller quantities. The finding that the amount of iron in some rocky planets is much higher than the amount in their host star would weaken the astronomers' claim because it would show that some planets contain the same material as their host star, but in higher quantities.

Choice A is incorrect because a finding only about the makeup of stars, whether they've cooled or not, would provide no information about the makeup of planets. Thus, it wouldn't have any bearing on the claim that planets and their host stars are composed of the same materials in differing quantities. Choice B is incorrect because a finding about two host stars having similar proportions of certain materials wouldn't provide any information about the makeup of planets. Thus, it wouldn't be relevant to the claim that planets and their host stars are composed of the same materials in differing quantities. Choice D is incorrect because the text indicates that the astronomers' claim is based on a fact—that stars and planets begin forming from the same gas and dust in space—which would remain true regardless of the effectiveness of a method for analysis of compositions. The text does cite analysis of rocky planets in our solar system and the Sun, but only as a single piece of evidence that is consistent with the claim and not as the source of the claim; the finding that the method used for that analysis is less effective in other scenarios wouldn't weaken a claim that's based on knowledge of how stars and planets initially form.

▼ 3.9 — C

Choice C is the best answer because it most effectively uses data from the graph to complete the statement about the researchers' hypothesis regarding a finding. The text states that the researchers attribute a certain finding about community assembly in fungi and prokaryotes to the fact that fungi have a higher tolerance for environmental pressures than do prokaryotes. This finding, then, would be explained by the fact that fungi better tolerate environmental pressures, which the text defines as including variable and homogeneous selection. According to the graph, the contribution of variable selection to community assembly was nearly 40 percent for fungi but nearly 60 percent for prokaryotes, indicating that variable selection contributed significantly less to fungal community assembly than to prokaryotic community assembly. This finding aligns with the researchers' hypothesis: because fungi tolerate environmental pressures like variable selection better

than prokaryotes do, those pressures play a smaller role in shaping fungal communities than in shaping prokaryotic communities.

Choice A is incorrect. Although this comparison is supported by the graph, which shows that homogenizing dispersal's contribution to fungal community assembly was just over 10 percent while variable selection's contribution was just under 40 percent, this comparison doesn't illustrate the hypothesis because it involves two factors affecting fungal community assembly. The researchers' hypothesis specifically concerns a difference between fungi and prokaryotes (namely, fungi's greater tolerance of environmental pressures), so a comparison that is limited to fungal community assembly doesn't represent a finding that aligns with this hypothesis. Choice B is incorrect because it mischaracterizes the information in the graph. The graph indicates that the total number of factors that contributed to community assembly was greater for fungi than for prokaryotes: all five factors made nonzero contributions to fungal community assembly, while two factors—homogeneous selection and dispersal limitation—appear not to have contributed to prokaryotic community assembly. In addition, the overall number of factors that influence community assembly isn't directly relevant to the researchers' hypothesis, which addresses the effects of higher tolerance of environmental pressures among fungi in comparison to prokaryotes. Choice D is incorrect. Although the graph indicates that homogeneous selection contributed only minimally to fungal community assembly and appears not to have contributed at all to prokaryotic community assembly, this isn't a finding that the researchers' hypothesis would account for. The hypothesis concerns a difference between fungi and prokaryotes that is plausibly attributable to fungi's greater tolerance of environmental pressures, which include homogeneous and variable selection. The finding that one type of environmental pressure contributed negligibly to both groups doesn't differentiate fungi from prokaryotes and therefore doesn't illustrate what the hypothesis is meant to explain.

▼ 3.10 — B

Choice B is the best answer because it correctly uses data from the table to illustrate the relationship between orbital and rotational periods described in the text. For Earth and each of four TRAPPIST-1 exoplanets, the table provides the orbital period and radius (in relation to the radius of Earth). The text explains that for each TRAPPIST-1 exoplanet, gravitational forces cause the rotational period (how long the planet takes to spin once on its axis) to equal the orbital period (how long the planet takes to orbit the star). According to the table, TRAPPIST-1e has an orbital period of 6.1 days. Since the text states that the orbital and rotational periods are equal for these planets, TRAPPIST-1e's rotational period must also be 6.1 days.

Choice A is incorrect because the table shows that TRAPPIST-1e has an orbital period (and thus a rotational period) of 6.1 days, not 1.51 days, which is the orbital period (and thus the rotational period) of TRAPPIST-1b. Furthermore, comparing the length of TRAPPIST-1e's rotational period to those of the other exoplanets would not logically complete the example, which should illustrate how the orbital and rotational periods of a planet in TRAPPIST-1 are the same length. Choice C is incorrect because the table shows that TRAPPIST-1e has an orbital period (and thus a rotational period) of 6.1 days, not 9.21 days, which the table shows is the orbital period (and thus rotational period) of TRAPPIST-1f. Additionally, comparing the rotational periods of two exoplanets would not logically complete the example, which should illustrate how the orbital and rotational periods of a planet in TRAPPIST-1 are the same length. Choice D is incorrect because although it correctly identifies TRAPPIST-1e's orbital period as 6.1 days, it incorrectly states that its rotational period is 0.92 days. This contradicts the text's statement that the orbital and rotational periods are equal for TRAPPIST-1 exoplanets. The value 0.92 actually represents TRAPPIST-1e's radius in relation to Earth's radius, not its rotational period.

▼ 3.11 — C

Choice C is the best answer because it most effectively completes the example regarding the ablation rate of iron. The table shows the ablation rates for three elements—iron, potassium, and sodium—found in cosmic dust that comes from one of four sources. The text says that the ablation rate for a given element in slower-moving SPC and AST dust was lower than the ablation rate for that same element in faster-moving HTC or OCC dust. The text then presents the first part of an example of this pattern, describing an ablation rate of 28% for iron in AST dust. The information that iron from HTC dust had an ablation rate of 90% is therefore the most effective way to complete this example—the comparison of a relatively low ablation rate for iron in slower-moving AST dust with a relatively high ablation rate for iron in faster-moving HTC dust illustrates the tendency of ablation rates for a given element to be lower in slower-moving dust than in faster-moving dust.

Choice A is incorrect because the text indicates that SPC dust, like AST dust, moves relatively slowly; a comparison of the ablation rates of iron from two slower-moving dust sources could not be an example of the difference between ablation rates in slower-moving dust and faster-moving dust, which is the pattern that the example is supposed to illustrate. Choice B is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation

rate of sodium. Choice D is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Additionally, any ablation rate from AST dust would be ineffective in this example since AST dust is referenced in the first part of the example and thus additional data focused on AST dust would not illustrate a variation across dust types.

▼ 3.12 — B

Choice B is the best answer. The finding that all the regions reported at least some effects on provisioning services supports the first part of the assertion. And the fact that provisioning services comprise 75% of the reported services overall supports the second part of the assertion.

Choice A is incorrect. This choice doesn't fully support the assertion. It doesn't include the finding that all the regions (not just Central and West) reported at least some effects on provisioning services. Choice C is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall. Choice D is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall.

▼ 3.13 — B

Choice B is the best answer. If there are parasites that feed on both snakes and frogs, they could carry *BovB* from snakes to frogs. HT could occur "indirectly" through those encounters, which frogs are more likely to survive than snake encounters.

Choice A is incorrect. The team's conclusion specifically argues that *BovB* is transmitted from snakes to frogs via HT, and this choice doesn't mention snakes. Choice C is incorrect. The team argues that *BovB* is transmitted from snakes to frogs via HT, and this answer choice discusses a scenario in which the gene is not transmitted. Choice D is incorrect. Whether or not frog species with *BovB* are advantaged isn't relevant to the team's conclusion about how the gene is transmitted.

▼ 3.14 — B

Choice B is the best answer because it presents a finding that, if true, would most directly support the idea advanced by Homero Gil de Zúñiga and Trevor Diehl that NFM ("news finds me") attitude may reduce voting probability through an indirect effect. The text describes NFM as an attitude that has lowered people's interest in actively acquiring news and introduces Gil de Zúñiga and Diehl's study on the effects of NFM on people in the United States' political knowledge and interest. The text goes on to say that despite the fact that the study didn't occur near a major election, Gil de Zúñiga and Diehl still conclude that NFM may reduce voting probability. If the likelihood, or probability, that a person will vote is linked to that person's level of political knowledge and interest, that would suggest that negatively affecting a person's level of political knowledge and interest would also negatively affect how likely that person is to vote. Thus, if NFM attitude has a negative effect on political knowledge and interest, then it would also likely reduce voting probability.

Choice A is incorrect because the finding that NFM attitude increases as major elections approach wouldn't address the effect of NFM on the likelihood of voting, which is the idea advanced by Gil de Zúñiga and Diehl. Moreover, although the text mentions that there were no major elections that occurred near the time of the study, it doesn't discern between major and minor elections when discussing voting probability. Choice C is incorrect because finding that NFM attitude shows little correlation with either political knowledge or political interest would undermine, not support, Gil de Zúñiga and Diehl's idea that NFM may reduce voting probability because it suggests that NFM has no effect on political knowledge or political interest. Choice D is incorrect. Although Gil de Zúñiga and Diehl's idea would be supported by the finding that the likelihood of voting increases as political knowledge increases, nothing in the text suggests that the researchers' idea hinges on the size of people's social networks.

▼ 3.15 — D

Choice D is the best answer because it most effectively uses a quotation from "Poetry" to illustrate the claim that the poem highlights an ambivalence, or a conflicted attitude, toward poetry. In the quotation, the speaker suggests that one might read poetry with "contempt," or disdain, for it, but even with this negative attitude one will find "a place for the genuine." Because the quotation expresses conflicting attitudes toward poetry, it effectively illustrates the speaker's ambivalence in discussing the merits and displeasure of reading poetry.

Choice A is incorrect because it doesn't mention poetry or show ambivalence. Choice B is incorrect. Although the idea of "half poets" may seem to relate to ambivalence, the speaker mentions only negative attitudes toward certain works and the quotation therefore lacks a

contrasting positive or neutral attitude that would be needed to indicate ambivalence. Choice C is incorrect because the speaker mentions only negative attitudes toward certain works and the quotation therefore lacks a contrasting positive or neutral attitude that would be needed to indicate ambivalence.

▼ 3.16 — D

Choice D is the best answer because it presents a statement that would support Halliwell's claim that "imitation" is an overly simplified translation of the word "mimesis" in the context of ancient Greek philosophical discussions of representational art and that, because it suggests a view of art as merely copying things that exist in reality, the translation obscures the fact that Plato, Aristotle, and other ancient Greeks understood the relationship between art and reality in varied ways. If Plato's works tend to treat representational art as an inferior or inadequate reflection of the physical world (consistent with the idea of mere imitation), while Aristotle's works suggest that the term "mimesis" can refer to art's ability to show conditions that don't currently exist but could exist (going beyond imitation into novel creation), that evidence would support the claim that the term "mimesis" means more than just "imitation" and that ancient Greek philosophers held different views of the relationship between art and reality.

Choice A is incorrect because information about the root word *mimos* first being used in drama and the root *mim-* coming to be associated with music and poetry wouldn't indicate anything about the meaning of the specific term "mimesis" as it was used by ancient Greek philosophers in discussions of representational art; thus, the information would have no bearing on Halliwell's claim that a common translation of the specific term is an oversimplification. Choice B is incorrect because the issue of art's psychological effects on audiences gets at how people respond to works of art instead of how art itself is related to reality, so the idea that Plato and Aristotle both addressed such effects in their considerations of mimesis wouldn't have any bearing on Halliwell's claim that a common translation of that term oversimplifies the relationship between art and reality. Further, this idea would touch on one general similarity in approaches to mimesis instead of illustrating the multifaceted, or varied, ways Halliwell claims ancient Greek philosophers understood art's relation to reality. Choice C is incorrect because the idea that Plato discussed aesthetics in his early works but didn't use the term "mimesis" until later doesn't give any indication of how Plato or any other ancient Greek philosophers understood the relation between art and reality, so it would have no bearing on Halliwell's claim that a common translation of the term "mimesis" is oversimplified and fails to reflect the varied ways the philosophers understood that relationship.

▼ 3.17 — A

Choice A is the best answer. If REM sleep were homeostatically regulated in fur seals, then all the seals would compensate with REM levels significantly over baseline after going weeks without REM. We'd also expect the seals to maintain those elevated REM levels for some time. Since seals B and C return very quickly to baseline REM levels, this suggests that REM sleep in fur seals may not be regulated homeostatically.

Choice B is incorrect. The graph doesn't depict REM sleep while in the water for the seals in the study. Additionally, we're told fur seals get no REM sleep while in the water, which is significantly different to the values shown in the graph for after they return to land. Choice C is incorrect. This doesn't support the conclusion. If REM sleep were homeostatically regulated in fur seals, then we'd suspect the seals to sustain REM levels well above baseline for a prolonged period in order to compensate for weeks of REM deprivation while in the water. Whether or not there's a reduction in REM sleep from day 1 to day 2 doesn't tell us how REM sleep on those days relates to baseline, which is where our focus should be. Choice D is incorrect. The y-axis of this graph doesn't depict baseline levels of REM sleep, but rather shows REM sleep as a percent of baseline.

▼ 3.18 — A

Choice A is the best answer because it most effectively uses data from the table to support the student's argument about the role of bufadienolide in the egg preferences of cane toad tadpoles. For each of five amphibian species included in the 2022 study, the table gives the percentage of available eggs that the cane toad tadpoles ate. According to the table, the tadpoles ate 10% of striped burrowing frog eggs and 1% of dainty green tree frog eggs, which suggests a preference for striped burrowing frog eggs over dainty green tree frog eggs. The table also indicates that neither of these species' eggs produces bufadienolide. Thus, these data suggest that something other than the presence or absence of bufadienolide is needed to adequately explain the tadpoles' egg preferences.

Choice B is incorrect. Although the table shows that for each of the five amphibian species, the cane toad tadpoles ate less than 100% of that species' eggs, which demonstrates that the tadpoles did indeed leave some eggs for each species unharmed, this fact alone is irrelevant to the tadpoles' preferences for some species' eggs over other species' eggs. Choice C is incorrect. Although the table indicates that the cane toad tadpoles ate 90% of the cane toad eggs and 7% of the short-footed frog eggs, which suggests that they prefer cane toad eggs over short-footed frog eggs, the table also indicates that cane toad eggs

produce bufadienolide, whereas short-footed frog eggs do not. Therefore, these data are not sufficient to exclude that bufadienolide alone could explain the tadpoles' preference for some species' eggs over other species' eggs. Choice D is incorrect. Although the table shows that for both dainty green tree frog eggs and little red tree frog eggs, the cane toad tadpoles ate 1% of those species' eggs, it also indicates that neither produces bufadienolide. Thus, these data alone don't indicate bufadienolide's role in the tadpoles' egg preferences.

▼ 3.19 — B

Choice B is the best answer because it describes data from the graph that support the researchers' conclusion that the trend of admirable-trait candidates being rated as more likable than ignoble-trait candidates held true when participants' own personality-trait scores were factored in, except among participants with high ignobility scores. The values on the x-axis represent survey participants grouped by their own ignobility scores, from low ignobility (1) to high ignobility (7), while the values on the y-axis represent the likability scores given to the political candidates. The graph shows that the full range of participants (from least to most ignoble) gave the admirable-trait candidates (represented by the line with triangles) a likability rating of approximately 70 out of 100; that is, regardless of their own level of ignobility, participants generally found admirable-trait candidates quite likable. However, the graph shows that participants varied in their views of ignoble-trait candidates (represented by the line with squares); likability ratings increased as the participants' own ignobility scores increased. Participants with low to medium-high ignobility scores (1 to 5) still rated the ignoble-trait candidates as less likable than the admirable-trait candidates, with all ratings falling below approximately 70, but participants with high ignobility scores (6 and 7) gave ratings equal to or higher than approximately 70. In other words, the previously observed trend of ranking admirable-trait candidates as more likable than ignoble-trait candidates persisted for participants with low to medium-high ignobility but not for participants with high ignobility.

Choice A is incorrect because it describes the opposite of what the graph shows. The graph shows a positive correlation between participants' ignobility scores and ignoble-trait candidates' likability ratings (as participants' ignobility scores increased, so did their ratings for ignoble-trait candidates' likability) and no correlation between ignobility scores and admirable-trait candidates' likability ratings (all participants gave admirable-trait candidates a rating of approximately 70 out of 100). Choice C is incorrect. The graph does show that regardless of their own ignobility scores, participants rated admirable-trait candidates as quite likable (a rating of approximately 70 out of 100). However, this doesn't support the

researchers' conclusion because the conclusion has to do with how participants rated both types of candidates, not just the admirable-trait ones; moreover, the conclusion is that relative ratings were actually affected by the participants' ignobility scores. Choice D is incorrect. The graph does show that only participants with an ignobility score of 6 gave the same likability score to both admirable- and ignoble-trait candidates while participants with other ignobility scores gave a different rating for each candidate, but this doesn't support the researchers' conclusion. The conclusion isn't just that participants gave different ratings to the two types of candidates—it's that participants with low to medium-high ignobility scores specifically gave higher likability ratings to admirable-trait candidates than to ignoble-trait candidates and that participants with high ignobility scores didn't.

▼ 3.20 — B

Choice B is the best answer because it presents a finding that, if true, would most directly support the arts journalist's claim about Enwezor's work as a curator and art historian. In the text, the arts journalist asserts that Enwezor wished not just to focus on modern African artists but also to show "how their work fits into the larger context of global modern art and art history," or how their work relates to artistic developments and work by other artists elsewhere in the world. The description of *Postwar: Art Between the Pacific and the Atlantic, 1945–1965* indicates that Enwezor and Siegel's exhibition brought works by African artists together with works by artists from other countries, thus supporting the arts journalist's claim that Enwezor sought to show works by African artists in a context of global modern art and art history.

Choice A is incorrect because it describes a retrospective that wouldn't support the arts journalist's claim that Enwezor wanted to show how works by modern African artists fit into the larger context of global modern art and art history. The description of *El Anatsui: Triumphant Scale* indicates that the retrospective focused only on the work of a single African artist, El Anatsui. The description doesn't suggest that the exhibition showed how El Anatsui's works fit into a global artistic context. Choice C is incorrect because it describes an exhibition that wouldn't support the arts journalist's claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* indicates that the exhibition showed how African artists were influenced by movements for independence from European colonial powers following the Second World War. Although this suggests that Enwezor intended the exhibition to place works by African artists in a political context, it doesn't indicate that the works were placed in a global artistic context. Choice D is incorrect because it describes an exhibition that wouldn't support the

arts journalist's claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *In/sight: African Photographers, 1940 to the Present* indicates that the exhibition was intended to reveal the broad range of approaches taken by African photographers, not that the exhibition showed how photography by African artists fits into a global artistic context.

▼ 3.21 — B

Choice B is the best answer because it presents a finding that, if true, would most directly support the researchers' conclusion that an observable pattern in time references in novels reflects a shift in human behavior prompted by the spread of electric lighting in the late nineteenth century. If novels published after 1880 contain significantly more references to activities occurring after 10 p.m. than novels from earlier periods do, this would suggest a change in human behavior and daily routines enabled by the availability of electric lighting. Before electric lighting—which provided illumination more easily than other available forms of light—many activities ceased after nightfall, so references to late-night activities would be less common in earlier novels. An increase in such references after 1880 would align with the researchers' conclusion, reflecting an increase in late-night activities made possible by electric lighting.

Choice A is incorrect because a decrease in references to 10 a.m. after the year 1800 would not support the researchers' conclusion involving a shift in human behavior prompted by the spread of electric lighting toward the end of the 1800s. The time of 10 a.m. is in the morning and, in most places, characterized by daylight, so a change in references to that time would not be clearly linked to the impact of electric lighting. Choice C is incorrect because while an increase in implied time references relative to clock phrases in nineteenth-century novels could suggest a change in writing style or conventions, it does not directly support the conclusion involving a shift in human behavior prompted by the spread of electric lighting. The text indicates that the researchers' conclusion is based on the content of the time references themselves, not the phrasing used. Choice D is incorrect. If references to noon and midnight are used with roughly the same frequency in all the novels analyzed by the researchers, this would reflect a lack of change in human behavior with regard to time and therefore would not support the researchers' conclusion involving a shift in human behavior that occurred in response to the spread of electric lighting.

▼ 3.22 — C

Choice C is the best answer because it presents a finding that, if true, would support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text explains that although archaeologists hold that categorizing pottery fragments relies on both objective criteria and instinct developed through direct experience, researchers have found that a computer model can categorize the fragments with the same degree of accuracy as the humans can—a finding that has caused some archaeologists to worry that their own work won't be needed any longer. If survey results indicate that categorizing pottery fragments limits the amount of time archaeologists can dedicate to other important tasks that only human experts can do, that would mean that computer models aren't able to do all of the important things archaeologists do, thus supporting the researchers' claim that computer models are unlikely to replace human archaeologists.

Choice A is incorrect because if it were true that the computer model could categorize the pottery fragments much more quickly than the archaeologists could, that would weaken the researchers' claim that archaeologists are unlikely to be replaced by certain computer models, since it would demonstrate that the models could conduct the archaeologists' work not only with equal accuracy but also at a faster pace. Choice B is incorrect because the inability of both the computer model and the archaeologists to accurately categorize all of the pottery fragments presented wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text indicates that some archaeologists are worried because the computer model's accuracy is equal to their own, and that could be the case whether both were perfectly accurate or were unable to achieve complete accuracy. Choice D is incorrect because survey results showing that few archaeologists received special training in properly categorizing pottery fragments wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The amount of special training in categorizing pottery fragments that archaeologists have received has no direct bearing on whether computer models' success at categorizing fragments will lead to the models replacing the archaeologists.

▼ 3.23 — B

Choice B is the best answer. This would weaken the explanation. If the recipients of electronic money immediately withdrew it all as physical money, then both kinds of recipients ended up spending physical money on food. So there must be some other explanation why those who initially received electronic money ate different kinds of food.

Choice A is incorrect. This wouldn't weaken the explanation. If anything, it actually supports it: it demonstrates that recipients of electronic money and recipients of physical money have

different spending habits. Choice C is incorrect. This wouldn't weaken the explanation. The explanation we're testing this choice against is about the way that people might "conceive of and allocate" physical and electronic funds differently. This choice only focuses on the timing, not what they spend the money on. Choice D is incorrect. This would have no impact on the explanation. It doesn't tell us anything about possible differences between the spending habits of those who spend physical money and those who spend money electronically.

▼ 3.24 — D

Choice D is the best answer because it presents a finding that, if true, would support Paredes's argument that Mexican-American folklore is mostly the result of cultural interactions in Mexico and the United States rather than an adaptation of Spanish folklore. The text describes a disagreement among scholars about whether Mexican-American folklore mostly derived from the folklore of Spain or originated in Mexico and the United States as cultures there have interacted. The latter view is the argument that Paredes puts forward. If Mexican-American folklore collected in the twentieth century mostly consists of ballads about history and social life that originated recently, then that would support Paredes's argument, since it would suggest that the folklore mostly arose after Spanish rule ended in the early nineteenth century and that the folklore reflects cultural interactions in Mexico and the United States rather than traditions from Spain.

Choice A is incorrect because the inclusion of songs influenced by sixteenth-century Spanish poetry among Mexican-American folklore collected in the twentieth century would not support Paredes's view that the folklore was the result of cultural interactions in Mexico and the United States rather than an offshoot of Spanish folklore. If anything, the presence of such songs among the folklore collected in the twentieth century would weaken Paredes's argument, since it would reflect the influence of Spanish culture on the folklore. Choice B is incorrect because the mere presence of similarities in Mexican-American folklore across regions would not be sufficient to draw a conclusion about where the folklore originated, let alone to support Paredes's argument that the folklore reflects cultural interactions in Mexico and the United States. In fact, since Paredes argued that Mexican-American folklore is the product of various cultures interacting in Mexico and the United States, he would likely expect there to be regional variations in folklore as different cultures have interacted in different places. Choice C is incorrect because scholars' previous ignorance of the folklore would have no bearing on Paredes's argument that Mexican-American folklore mostly reflects cultural interactions in Mexico and the United States. The folklore's origins are independent of scholars' knowledge of the folklore.

▼ 3.25 — D

Choice D is the best answer. This critic challenges the claim by arguing that *A History of Mexico* is not so idiosyncratic (unique) as to resist classification because its use of Maya-style human profiles actually "invites classification as Indigenous art." Therefore, according to this critic, the work can be viewed, at least partially, through a lens of national or cultural identity.

Choice A is incorrect. While it describes the Maya influence on a particular aspect of *A History of Mexico*, this quotation doesn't push back on Medellín's "idiosyncratic" (unique) style, nor does it try to categorize the entire work into a single cultural tradition. Choice B is incorrect. This choice doesn't directly challenge the underlined claim, but rather supports it. It suggests that the work effectively blends a variety of artistic products to create a single work that can't be defined by any one tradition. Instead, the work recalls many centuries of history and culture all at once. Choice C is incorrect. This choice supports rather than challenges the underlined claim. The quotation argues that other critics focus too much on Indigenous influences on the artwork instead of viewing all of the influences equally—that they should instead be viewing the work as an idiosyncratic whole instead of through one or two narrow cultural lenses.

▼ 3.26 — C

Choice C is the best answer because it uses data from the graph to effectively complete the example of Eludoyin and his colleagues' findings concerning female farmers in some regions of Ondo State, Nigeria. The graph presents values for the percentage of Ondo State small-scale farmers who are female, by type of crop and region. The graph shows that of the farmers mainly cultivating non-root vegetables, approximately 57% in north Ondo and approximately 54% in south Ondo are female; in other words, most of those farmers are female, which exemplifies the idea that female farmers make up the majority (more than half) of the farmers cultivating specific types of crops in some regions.

Choice A is incorrect because it inaccurately cites data from the graph: the graph shows that in south Ondo, most of the farmers mainly cultivating non-root vegetables are women (approximately 54%), but that only about 35% (less than half) of the farmers mainly cultivating cereals are women. Choice B is incorrect because it inaccurately cites data from the graph: the graph shows that more women in central Ondo mainly cultivate cereals than mainly cultivate root crops (approximately 36% and 20%, respectively). Additionally, it doesn't effectively complete the example because the graph shows that female farmers don't make

up the majority (more than half) of the farmers for any type of crop in central Ondo. Choice D is incorrect because it doesn't effectively complete the example; it simply states that a relatively equal proportion of women across the three regions mainly cultivate cereals, which doesn't address the value for that proportion and thus doesn't show that a majority (more than half) of the farmers cultivating certain crops are female.

▼ 3.27 — C

Choice C is the best answer because it effectively completes the text's discussion of Moses and colleagues' analysis of changes in Neptune's temperature by using data from the table to identify the source of the earliest example of spectroscopy data included in this analysis. The text indicates that Moses and colleagues analyzed infrared imaging and spectroscopy data about Neptune obtained in different years by various instruments. The table lists the observatories where these instruments are housed, the type of data each instrument collected (infrared imaging or spectroscopy), and the years in which the instruments' observations were made. According to the table, only two of these instruments obtained spectroscopy data: VISIR at the European Southern Observatory, which made its observations in 2006, and TEXES at the Gemini Observatory, which made its observations in later years (2007 and 2019). Although two other instruments are also included in the table, they obtained infrared imaging data, not spectroscopy data. Therefore, the earliest example of spectroscopy data included in the researchers' analysis is the data obtained by VISIR at the European Southern Observatory in 2006.

Choice A is incorrect. Although the table indicates that TEXES at the Gemini Observatory collected spectroscopy data in 2007 and 2019, the table also indicates that this type of data was collected in 2006 by VISIR at the European Southern Observatory. Because the VISIR data were collected in the previous year, the data obtained in 2007 by TEXES are not the earliest example of spectroscopy data included in Moses and team's analysis of Neptune's temperature. Choice B is incorrect because the table indicates that T-ReCS at the Gemini Observatory collected infrared imaging data, not spectroscopy data. T-ReCS therefore cannot be the source of the earliest example of spectroscopy data that is included in Moses and colleagues' analysis of Neptune's temperature. Additionally, the T-ReCS data were collected in 2007, meaning that these data were obtained later than any type of data collected in 2006. Choice D is incorrect. Although the table indicates that data from LWS at the W.M. Keck Observatory were obtained in 2003, making these data the earliest example of any type of data included in the analysis, the table also indicates that LWS collected infrared imaging data, not spectroscopy data. LWS therefore cannot be the source of the

earliest example of spectroscopy data included in Moses and team's analysis of Neptune's temperature.

▼ 3.28 — D

Choice D is the best answer because it describes data from the graph that support Ibáñez and colleagues' conclusion that increasing anthropogenic nitrogen deposition can compensate for the negative effect of climate change on tree growth if that change is moderate but not if it's extreme. The bar graph shows the growth of sugar maple trees with and without nitrogen fertilization under three different climate-change scenarios: current conditions, a moderate change, and an extreme change. According to the graph, radial growth without nitrogen fertilization is projected to be about 0.16 centimeters (cm) under current conditions, 0.15 cm under a moderate change, and 0.04 cm under an extreme change. The graph also shows that with nitrogen fertilization, growth is projected to be about 0.18 centimeters under a moderate change but only about 0.06 centimeters under an extreme change. Thus, the data in the graph support the researchers' conclusion by showing greater growth for a moderate change using nitrogen fertilization than they do either under current conditions without nitrogen fertilization or under an extreme change with nitrogen fertilization.

Choice A is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of anthropogenic nitrogen, but only if change to the climate is moderate and not if it's extreme. To support this would require comparing radial growth without nitrogen fertilization under current climate conditions to the growth with nitrogen fertilization under both moderate and extreme changes. This choice mentions only growth with nitrogen fertilization under current climate conditions and moderate change and growth without nitrogen fertilization under an extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change. Choice B is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of atmospheric nitrogen, but only if change to the climate is moderate and not if it's extreme. The support needed would compare radial growth under current climate conditions without nitrogen fertilization to the growth with nitrogen fertilization under moderate and extreme changes. This choice mentions only growth without nitrogen fertilization under current conditions and moderate change and growth with nitrogen fertilization under extreme change, which don't provide a basis to determine whether higher

nitrogen in the future will be able to offset reduced growth due to climate change. Choice C is incorrect. Although it accurately represents the data in the graph, this fact pattern doesn't support Ibáñez and colleagues' conclusion that the decline in radial growth due to climate change will be partly offset by higher levels of atmospheric nitrogen, but only if change to the climate is moderate and not if it's extreme. The support needed would compare radial growth without adding nitrogen under current climate conditions to the growth with nitrogen fertilization under moderate and extreme changes. This choice mentions only the growth with and without nitrogen fertilization under moderate climate change and growth without nitrogen fertilization under extreme change, which don't provide a basis to determine whether higher nitrogen in the future will be able to offset reduced growth due to climate change.

▼ 3.29 — A

Choice A is the best answer because it presents a conclusion about the habits of New Year's resolution makers that is best supported by information in the text and the table. The table presents percentages of resolution makers who make particular kinds of resolutions, indexed by age bracket. According to the table, 58 percent of resolution makers ages 50 to 64 made resolutions related to personal relationships, compared to 53 percent of resolution makers ages 30 to 49—indicating that the older group is more likely to make such resolutions. The table also shows that 56 percent of resolution makers ages 50 to 64 made resolutions related to finances, compared to 63 percent of resolution makers ages 30 to 49—indicating that the older group is less likely to make financial resolutions. Thus, the data support the conclusion that resolution makers between the ages of 50 and 64 are more likely to make resolutions related to personal relationships and less likely to make resolutions related to finances than resolution makers between the ages of 30 and 49 are.

Choice B is incorrect because according to the table, 79 percent of resolution makers ages 18 to 29 made resolutions about health and exercise, while 80 percent of resolution makers ages 30 to 49 made such resolutions. Therefore, resolution makers ages 18 to 29 are slightly less likely, not more likely, to make resolutions about health and exercise than resolution makers ages 30 to 49 are. Choice C is incorrect because neither the text nor the table provides information about whether resolution makers who make resolutions related to health and exercise also make resolutions in multiple additional categories. The text indicates that half of those who make a resolution make more than one, but it doesn't specify how many or in which categories, and the table presents only the percentages of resolution makers in each age bracket who made particular kinds of resolutions. Choice D is incorrect because the table presents data only about people who chose to make resolutions, not about all US adults; the table therefore can't support a conclusion about whether US

adults overall become less likely to make resolutions as they age. Furthermore, although percentages of resolution makers generally decline as age bracket increases in some resolution categories, the table shows that the percentage of resolution makers who made health and exercise resolutions remains relatively stable across age brackets; thus, even among resolution makers, there isn't a consistent age-related decline across all types of resolutions.

▼ 3.30 — D

Choice D is the best answer because it describes data from the graph that support the researchers' conclusion that there is a growing interest among CEOs in connecting with more departments in their companies. The graph shows the average number of individuals reporting directly to CEOs during three different time periods: the individuals are divided into managers and department leaders. The average number of department leaders directly reporting to their CEO during the 1991–1995 period was slightly more than three, during the 1996–2001 period it was four, and during the 2001–2008 period it was almost seven. Thus, the average number of department leaders reporting directly to their CEO rose over the three periods studied, which suggests that CEOs were connecting with more departments.

Choice A is incorrect because the average number of managers and department leaders reporting directly to their CEO rose for both categories between the 1991–1995 and 2001–2008 periods; thus, it isn't true that the average numbers didn't fluctuate. Choice B is incorrect because the average number of managers reporting directly to their CEO was highest in the 2001–2008 period, not in the 1996–2001 period. Choice C is incorrect. Although it correctly describes a feature of the graph, the observation that more department leaders than managers are reporting to CEOs does not by itself address the question of whether CEOs are connecting with more departments over time—to address that question, one needs to know whether the number of department leaders reporting to CEOs is increasing over time.

▼ 3.31 — D

Choice D is the best answer because it uses data from the graph to effectively illustrate the text's claim about general economic policy uncertainty in the United Kingdom. The graph presents values for economic policy uncertainty in tax and public spending policy, trade policy, and general economic policy in the UK from 2005 to 2010. The graph shows that in 2005, the value for general economic policy uncertainty (approximately 90) was substantially lower than the value for uncertainty about trade policy specifically (approximately 160). It

also shows that in 2010, the value for general economic policy uncertainty (approximately 120) was substantially higher than the value for uncertainty about trade policy (approximately 70). The substantial differences between these values in 2005 and 2010 support the claim that a general measure may not fully reflect uncertainty about specific areas of policy.

Choice A is incorrect because the graph shows that the level of general economic policy uncertainty was similar to the level of uncertainty about tax and public spending policy in both 2005 (with values of approximately 90 and 100, respectively) and 2009 (with values of approximately 80 and 75, respectively). Choice B is incorrect because the graph shows that general economic policy uncertainty was higher than uncertainty about tax and public spending policy in 2006, 2007, and 2009, not that it was lower each year from 2005 to 2010. Choice C is incorrect because the graph shows that general economic policy uncertainty reached its highest level in 2010, which was when uncertainty about tax and public spending policy also reached its highest level, not its lowest level.

▼ 3.32 — B

Choice B is the best answer because the quotation from *The Souls of Black Folk* illustrates the claim that Du Bois felt a sense of cultural recognition when he heard Black folk songs. In the quotation, Du Bois explains that for his entire life, Black folk songs "stirred [him] strangely." Even though they originated in the South, a region he wasn't familiar with, he knew the songs "as of me and of mine." That is, he identified strongly with them and associated them with his community. Therefore, Du Bois felt an intuitive sense of cultural recognition when he heard Black folk songs.

Choice A is incorrect. Although the quotation considers the cultural and spiritual value of Black folk music, it doesn't establish that this music inspired in Du Bois a sense of cultural recognition. Choice C is incorrect because this quotation addresses the cultural survival of Black folk songs despite attempts to caricature, or parody, them, not Du Bois's sense of cultural connection to them. Choice D is incorrect because the quotation indicates that the Black folk songs and music are old, "the siftings of centuries," instead of addressing how Du Bois felt when he heard the songs.

▼ 3.33 — C

Choice C is the best answer because it most effectively uses a quotation from *King Lear* to illustrate the claim that King Lear expresses regret for his actions. In the quotation, Lear describes striking himself on the head—the same act he's engaged in as he speaks, and

one that suggests he's deeply upset with himself. Referring to himself in the second person (with "thy"), the character exclaims "Beat at this gate that let thy folly in / And thy dear judgement out!" Lear refers metaphorically to his own mind as a gate that has allowed folly, or poor judgement, to enter and good judgement to escape. This suggests that Lear regrets his attempts to test his three daughters' devotion to him, regarding those attempts as examples of the folly that has entered the gate of his mind.

Choice A is incorrect because this quotation doesn't express King Lear's sense of regret over his own actions; instead, it expresses his belief that the harm that others have done to him (or the extent to which they have "sinned against" him) outweighs whatever harm he himself has caused by "sinning." Choice B is incorrect because this quotation doesn't express King Lear's sense of regret over his own actions; instead, it expresses his thoughts about an approaching storm ("this tempest"), which he believes "will not give [him] leave to ponder," or time to consider, the harm that he will continue to experience ("things" that "would hurt [him] more"). Choice D is incorrect because this quotation expresses King Lear's vow to commit terrible actions (or "things" that "shall be / The terrors of the earth") in the future, not his regret over actions that he's already taken.

▼ 3.34 — B

Choice B is the best answer because it describes data from the graph that complete the text's discussion of lizard species' use of maximal speed when escaping predators. According to the text, moving at maximal speed (the highest speed possible) requires so much energy that it is not always an effective strategy for animals, even when they are escaping predators. The graph displays data on the average percent of maximal speed used by lizard species while either escaping predators or pursuing prey. The graph categorizes the data for both pursuing and escaping by the number of species using 30%–39% of maximal speed, 40%–49% of maximal speed, 50%–59% of maximal speed, 60%–69% of maximal speed, 70%–79% of maximal speed, 80%–89% of maximal speed, and 90%–100% of maximal speed, respectively. In the graph, there is at least one species in each of the following percent categories for maximal speed while escaping predators: 50%–59%, 60%–69%, 70%–79%, and 80%–89%. Thus, the data in the graph show that multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.

Choice A is incorrect because the data in the graph isn't organized in such a way that a comparison of the percentage of maximal speed used when escaping predation with the percentage used when pursuing prey is possible at the level of individual species. Choice C

is incorrect. It is true that in the graph, the percent category with the largest number of species using maximal speed while escaping predators is 90%–100% (8 species total). However, these data don't complete the text, which is concerned instead with how animals are discouraged from using maximal speed even when escaping predators because of the amount of energy required to use it. Choice D is incorrect because these data from the graph pertain to maximal speed while pursuing prey and therefore don't complete the text's discussion of lizard species' use of maximal speed when escaping predators.

▼ 3.35 — B

Choice B is the best answer because it presents a finding that, if true, would most directly weaken the conclusion that Wang and colleagues' findings indicate a trade link between Óc Eo and South Asia. The text explains that turmeric (native to South Asia) and footed grinding slabs of a style common in South Asia between roughly 500 BCE and 300 CE were found at Óc Eo, a Southeast Asian port that flourished between the first and sixth centuries CE. From this evidence, the text concludes that these items must have reached Óc Eo through trade. If, however, there was a significant migration of people from South Asia to Southeast Asia during the first through third centuries CE, that would offer an alternative explanation: South Asian people could have brought turmeric and grinding slabs with them when they migrated. Thus, the finding would weaken the conclusion that trade must account for the items' presence at Óc Eo.

Choice A is incorrect. A finding involving spices from a different region—the Maluku Islands, southeast of Óc Eo—wouldn't undermine the conclusion because it wouldn't reveal how the items from South Asia, a region to the west, reached Óc Eo. If anything, such a finding might be further evidence of broad trade networks involving Óc Eo. Choice C is incorrect. If it were true that other South Asian artifacts from the period when Óc Eo flourished have been found throughout Southeast Asia, that would suggest only that South Asian items other than turmeric and footed grinding slabs were present in the region at the time; it wouldn't reveal anything about how any of those items reached Southeast Asia from South Asia. If anything, the widespread presence of South Asian artifacts in Southeast Asia during this period could be seen as further support for a trade network connecting the regions. Choice D is incorrect because the finding that Óc Eo and some South Asian communities traded with people in another Southeast Asian location (what is now Malaysia) would suggest that this location could have been an intermediary for the movement of items from South Asia to Óc Eo. Such a finding would support, not weaken, the plausibility of an indirect trade link that's presented in the conclusion.

▼ 3.36 — B

Choice B is the best answer. This choice weakens the student's conclusion by suggesting that the emotionality of movie reviews is related to box office success: the higher the emotionality score, the better the movie performs at the box office.

Choice A is incorrect. While this choice does mention emotionality scores, it only connects them to star ratings, not to box office success. Choice C is incorrect. The fact that many movies received an average star rating doesn't tell us anything about the relationship between emotionality and box office success. Choice D is incorrect. While this choice suggests that star ratings can predict box office success, it doesn't address the issue of emotionality in written reviews, which is the focus of the student's conclusion.

▼ 3.37 — B

Choice B is the best answer because it effectively uses data from the graph to complete the statement about Rodrigo da Costa Portilho-Ramos and colleagues' conclusion. The graph shows the ratio of manganese to calcium in *L. pertusa* coral samples from the Alboran Sea and the Mauritanian coast. The graph reflects time in approximate years before present: in other words, the greater the number in years noted on the graph's horizontal axis, the farther that moment is in the past. The text indicates that the researchers tested the samples to determine whether oxygenation played a role in the decline of *L. pertusa*. The text goes on to note that a change in the ratio of manganese to calcium would signal an inverse, or opposite, change in oxygenation. According to the graph, the ratio of manganese to calcium in samples from the Alboran Sea increased from about 30 micromoles per mole 10,000 years ago to about 80 micromoles per mole 8,000 years ago, which means that oxygenation decreased between 10,000 and 8,000 years ago. Meanwhile, there was almost no discernible change in the ratio of manganese to calcium in samples from the Mauritanian coast between 12,000 and 10,000 years ago. According to the text, the population of *L. pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Thus, the increase in the ratio of manganese to calcium around 9,000 years ago in the Alboran Sea coincides with the decline in the *L. pertusa* population, suggesting an association between the decrease in oxygenation and the decline in population of the coral. No such relationship is suggested around 11,000 years ago near the Mauritanian coast. So, oxygenation likely played a role in the *L. pertusa* decline in the Alboran Sea but not in the coral's decline near the Mauritanian coast.

Choice A is incorrect because it asserts the opposite of what the graph indicates regarding oxygenation in the Alboran Sea, and it misrepresents what the graph indicates about oxygenation near the Mauritanian coast. The graph indicates that at the time of the decline in *L. pertusa* (approximately 9,000 years ago), the samples from the Alboran Sea contained a ratio of manganese to calcium that was increasing. According to the text, this ratio inversely correlates with ocean oxygenation levels, so if the ratio was increasing, oxygenation was decreasing, not substantially increasing. Furthermore, the graph shows that the ratio of manganese to calcium remained relatively stable in coral samples from the Mauritanian coast during the period studied, which suggests that there was no discernible relationship between oxygenation and the coral's population decline in that location, not that there was a substantial decrease in oxygenation corresponding to the coral's decline. Choice C is incorrect. Although the graph suggests that the level of oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after—because the ratio of manganese to calcium inversely correlates with ocean oxygenation levels and this ratio was lower before the decline than after—the graph doesn't support the claim that oxygenation near the Mauritanian coast was consistently low before and after the coral's decline there. Rather, the graph indicates that relative to coral samples from the Alboran Sea, the ratio of manganese to calcium in samples from near the Mauritanian coast was consistently low, which suggests that oxygenation levels were relatively high both before and after the decline of *L. pertusa*. Choice D is incorrect because it states the opposite of what the graph indicates: the graph shows that throughout the period studied, the ratio of manganese to calcium was higher in coral samples from the Alboran Sea than it was in samples from near the Mauritanian coast. Since the text indicates that the ratio of manganese to calcium inversely correlates with ocean oxygenation levels, oxygenation in the Alboran Sea was therefore lower than, not higher than, oxygenation near the Mauritanian coast during the period studied. Moreover, even if choice D did accurately represent the graph, it wouldn't effectively complete the statement since a comparison of the ocean oxygenation levels at the two locations is not relevant to the claim that a decline in oxygenation levels was associated with the decline of *L. pertusa* in the Alboran Sea but not near the Mauritanian coast.

▼ 3.38 — C

Choice C is the best answer because it most effectively uses data from the graph to complete the text about the effect of messaging on participative pricing. The graph shows mean ticket prices chosen by participants in response to three messages across two studies: Study 1, which the text indicates was conducted with an age-diverse group recruited

online, and Study 2, which was conducted with student participants. The graph indicates that in the "pay what you think it's worth" condition, the mean price of the concert tickets in Study 2 was about $74, which is greater than the mean price of about $55 in Study 1. In other words, when participants were asked to consider their valuation of the tickets, the response was heterogeneous, or mixed. Moreover, according to the graph, both Study 1 and Study 2 show higher prices for the tickets under the "pay what you think it's worth" condition than they do under both the "pay what you can" and the "pay what you want" conditions. That is, the data suggest that both groups of participants named higher prices when considering the value of the tickets than when considering either what they could afford or wanted to pay, a finding that supports the idea that sellers can benefit when prompting consumers to consider their own valuations when they choose prices.

Choice A is incorrect because it contradicts information in the graph. Although the graph shows that students in Study 2 assigned a higher value to the tickets than did the age-diverse group in Study 1, which would support the idea that consumer valuations were heterogeneous, the graph shows that in the "pay what you can" (i.e., what you can afford) condition, the students in Study 2 assigned a higher price (about $40), not a lower price, than the age-diverse group in Study 1 did (about $30). Moreover, even if it were true that the students had assigned a lower price in this condition, it wouldn't support the result described in the text, only that the participants across the two studies had different ideas of what they can afford to pay. Choice B is incorrect. Although a finding that participants tended to choose prices that were closest to the actual ticket costs in the "pay what you think it's worth" condition would support the idea that sellers benefit by prompting consumers to think about their own valuations (since it's implied that sellers would lose money in the other conditions, where chosen prices were lower than the participants' valuations), neither the text nor the graph addresses how any of the prices chosen by the study participants relate to the tickets' actual market price. Choice D is incorrect. Although the wide variation in participant valuations would support the idea that consumer valuations tend to be heterogeneous, neither the text nor the graph provides any information from which to discern the relative levels of variance among the responses from participants in either study.

▼ 3.39 — A

Choice A is the best answer because it accurately uses data from the table to complete the statement about the fibers of three types of sheep. The text discusses some qualities of outer and inner fibers of mouflons, Spanish Merino sheep, and Navajo-Churro sheep and indicates that sheep fibers with finer, or smaller, diameters are softer than fibers with larger diameters. The text then sets up a conclusion that can be made by comparing the fibers of

all three sheep types. The table shows that the minimum diameter of the Navajo-Churro's inner fibers is 10 microns, which is smaller than the minimum diameters of the inner fibers of both the mouflon (15 microns) and the Spanish Merino (17 microns). Thus, comparing the fiber characteristics of all three sheep reveals that at its softest, the inner fiber of the Navajo-Churro is softer than either the mouflon's or the Merino's inner fiber.

Choice B is incorrect because it doesn't accurately describe the data in the table. The table shows that the softness of Navajo-Churro inner fiber isn't uniform (since the diameter of Navajo-Churro inner fiber ranges from 10 microns to 35 microns) and that the diameter of the outer fiber of the mouflon isn't variable (rather, it is a uniform 150 microns). Choice C is incorrect because it doesn't accurately describe the data in the table. The text indicates that both Spanish Merino sheep and Navajo-Churro sheep were domesticated from mouflon sheep. The table doesn't show that domestication increased the minimum diameter of inner fibers for both Spanish Merino sheep and Navajo-Churro sheep from the 15-micron diameter of mouflon inner fibers: the minimum diameter of Spanish Merino increased to 17 microns, but the minimum diameter of Navajo-Churro inner fibers *decreased* to 10 microns. Choice D is incorrect because it doesn't accurately describe the data in the table. The text explains that sheep fibers with finer, or smaller, diameters are softer than fibers with larger diameters. The table does show that Spanish Merino outer fibers are finer, and thus softer, than mouflon outer fibers and that at their finest, Navajo-Churro outer fibers are softer than mouflon outer fibers, but it also shows that Spanish Merino inner fibers are less soft than mouflon inner fibers, since their minimum diameter is greater than 15 microns, and that Navajo-Churro inner fibers can be less soft than mouflon inner fibers, since the diameter ranges up to 35 microns.

▼ 3.40 — D

Choice D is the best answer because it presents a finding that, if true, would most directly support Barboza and Trejos's claim that rural female entrepreneurs who have received small loans from ALSOL are strategic in selling their goods less frequently than they could, even if it means missing payments. The text explains that borrowers in the ALSOL program use proceeds from their businesses to repay loans in equal weekly payments, with almost no penalty for missed payments. According to the text, Barboza and Trejos found that rural borrowers miss weekly payments in part because they don't sell their goods as often as they could, a move the researchers claim allows the entrepreneurs to help increase profits for the goods they sell. Finding that the cost of bringing goods to towns with marketplaces is high for rural entrepreneurs but is largely independent of how many goods are brought would support the researchers' claim: traveling to marketplaces less frequently would mean that a

rural entrepreneur spends less on travel overall, and taking a large load of goods to a marketplace for essentially the same cost as taking a small load would allow the entrepreneur to more substantially offset the cost of travel with greater overall sales at the marketplace, resulting in more profit per good sold—even if those profits are earned less frequently and don't support weekly loan payments.

Choice A is incorrect because the finding that many marketplaces require entrepreneurs to pay the operators of the marketplace a fixed percentage of proceeds to be able to sell goods there wouldn't explain why rural entrepreneurs strategically choose to sell their goods less frequently than they could in order to increase their profits per unit sold. With a fixed percentage of proceeds due to operators, the amount entrepreneurs have to pay operators would also be fixed regardless of frequency of selling. Choice B is incorrect because the finding that rural entrepreneurs can usually sell their goods for higher prices in cities than in their local areas but also face higher competition to sell goods in cities wouldn't explain why rural entrepreneurs strategically choose to sell their goods less frequently than they could in order to increase their profits per unit sold. This is because both the higher prices and higher competition in cities would be stable factors—meaning there would be no clear reason for the rural entrepreneurs not to take every available chance to sell their goods in cities and to instead sell their goods in cities only sometimes. Choice C is incorrect because the finding that rural entrepreneurs have lower costs and thus tend to require smaller initial loans than urban entrepreneurs do has no bearing on rural borrowers strategically choosing to sell their goods less frequently than they could specifically to increase their profits per unit sold. The cost of producing goods doesn't depend on the frequency with which an entrepreneur sells those goods, so lower frequency alone wouldn't affect profits, and the initial loan amount is set and has nothing to do with how much profit is earned from each sale.

▼ 3.41 — B

Choice B is the best answer because it presents a finding that, if true, would best account for the discrepancy Etta-Nkwelle observed: Asian countries showed evidence of herding (converging on the same foreign currency accumulation rate as their regional peers), while African countries did not. The text defines herding as individuals converging on the same action by imitating their peers, which implies that the individuals choose to do so rather than being compelled to follow a particular course of action. This choice indicates that most of the African countries belonged to economic unions that dictated such monetary policies, whereas Asian countries were free to choose, and the similarity among the Asian countries

suggests that herding can account for that similarity. The finding thus identifies a relevant difference between the two regions that accounts for the discrepancy.

Choice A is incorrect. Although this choice is related to reserves of foreign currency, the fact that African and Asian countries had different mixes of foreign and domestic currency reserves does not address the rate of foreign currency accumulation or whether herding is a contributing factor determining the rate of accumulation. Choice C is incorrect. This choice does not address currency accumulation. And although it describes differences in the currency reserves of African and Asian countries shortly after the 1997 financial crisis, it does not address the rate of foreign currency accumulation nor how similar the nations were in this regard to their neighboring countries. Choice D is incorrect. Although this choice discusses the variety of foreign currencies accumulated by African countries as compared with Asian countries, it does not discuss the rate of foreign currency accumulation nor whether herding is a factor determining the rate of accumulation.

▼ 3.42 — B

Choice B is the best answer because it presents a finding that would best support Granito and Álvarez's claim that fermenting black beans makes them easier to digest and more nutritious. The text indicates that high levels of soluble fiber and raffinose in black beans make the beans hard to digest and that tannins and trypsin inhibitors make it harder for the body to extract nutrients from the beans. If it were found that fermenting the beans significantly reduces their levels of soluble fiber, raffinose, trypsin inhibitors, and tannins when cooked, this would directly support the claim that fermentation improves the digestibility of the beans and makes them more nutritious.

Choice A is incorrect because the text indicates that trypsin inhibitors and tannins interfere with the body's ability to extract nutrients from black beans; if fermentation and cooking were found to increase these antinutrients, fermented beans would likely be less nutritious than unfermented ones, not more nutritious (as Granito and Álvarez claim). Choice C is incorrect because the text doesn't address the idea that greater nitrogen absorption in the gut has an effect on a food's digestibility or level of nutrition, so the discovery of the presence of microorganisms that may increase nitrogen absorption wouldn't provide relevant support for the claim that fermentation makes black beans easier to digest and more nutritious. Choice D is incorrect because Granito and Álvarez's claim focuses on the effect of fermenting black beans, but the finding that nonfermented black beans also have fewer trypsin inhibitors and tannins when cooked at high pressure would suggest that the role of

the cooking method could be significant when it comes to nutrition; further, the finding wouldn't address the beans' digestibility.

▼ 3.43 — D

Choice D is the best answer because it presents a finding that, if true, would weaken the scientists' hypothesis about icebergs that appear to be green. The text indicates that most icebergs are either mostly white or blue in color but that some icebergs in Antarctica appear to be green. The text goes on to say that the scientists hypothesized that this green color occurs when yellow-tinted dissolved organic carbon in ocean waters mixes with blue ice. A finding that both blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon would suggest that something other than yellow-tinted organic carbon causes some icebergs' green color, since the blue icebergs that contain yellow-tinted organic carbon remained blue instead of turning green.

Choice A is incorrect because, according to the text, the scientists' hypothesis was that blue icebergs, not white ones, change color when their ice mixes with yellow-tinted dissolved organic carbon. A finding that white ice, because of its air bubbles, doesn't change color when it's mixed with dissolved organic carbon would therefore have no bearing on the scientists' hypothesis. Choice B is incorrect because the text focuses only on Antarctic icebergs that appear to be green. It doesn't indicate that icebergs in locations other than Antarctica have been found to have a green hue. A finding that dissolved organic carbon has a stronger yellow color in Antarctic waters than in other places would therefore have no bearing on the scientists' hypothesis that green color in icebergs in Antarctica is caused by yellow-tinted dissolved organic carbon mixing with blue ice. Choice C is incorrect because, according to the text, the scientists' hypothesis was that blue icebergs turn green when their ice mixes with yellow-tinted dissolved organic carbon in the sea around them. If that's correct, one would expect blue icebergs and green icebergs to be located at a distance from each other since all blue icebergs in an area where the waters contain yellow-tinted dissolved organic carbon would take on a green hue. A finding that blue icebergs and green icebergs are rarely found near each other would therefore strengthen, not weaken, the researchers' hypothesis.

▼ 3.44 — C

Choice C is the best answer because it presents a finding that, if true, would weaken Foster's hypothesis that damage to eelgrass roots improves the health of eelgrass meadows by boosting genetic diversity. The text indicates that sea otters damage eelgrass roots but

that eelgrass meadows near Vancouver Island, where there's a large otter population, are comparatively healthy. When Foster and her colleagues compared the Vancouver Island eelgrass meadows to those that don't have established otter populations, the researchers found that the Vancouver Island meadows are more genetically diverse than the other meadows are. This finding led Foster to hypothesize that damage to the eelgrass roots encourages eelgrass reproduction, thereby improving genetic diversity and the health of the meadows. If, however, other meadows not included in the study are less healthy the larger the local otter population is and the longer the otters have been in residence, that would suggest that damage to the eelgrass roots, which would be expected to increase with the size and residential duration of the otter population, isn't leading meadows to be healthier. Such a finding would therefore weaken Foster's hypothesis.

Choice A is incorrect because finding that small, recently introduced otter populations are near other eelgrass meadows in the study wouldn't weaken Foster's hypothesis. If otter populations were small and only recently established, they wouldn't be expected to have caused much damage to eelgrass roots, so even if those eelgrass meadows were less healthy than the Vancouver Island meadows, that wouldn't undermine Foster's hypothesis. In fact, it would be consistent with Foster's hypothesis since it would suggest that the greater damage caused by larger, more established otter populations is associated with healthier meadows. Choice B is incorrect because the existence of areas with otters but without eelgrass meadows wouldn't reveal anything about whether the damage that otters cause to eelgrass roots ultimately benefits eelgrass meadows. Choice D is incorrect because the health of plants other than eelgrass would have no bearing on Foster's hypothesis that damage to eelgrass roots leads to greater genetic diversity and meadow health. It would be possible for otters to have a negative effect on other plants while nevertheless improving the health of eelgrass meadows by damaging eelgrass roots.

▼ 3.45 — A

Choice A is the best answer because it presents a finding that, if true, would support the text's conclusion about the results of the experiment with Tiffany & Co. and Forever 21. According to the text, Sevilla and Townsend found in several experiments that when products in a generic retail setting are displayed with a low product-to-space ratio (that is, with lots of space between them), consumers view those products as significantly more valuable than when the same products are displayed with a higher product-to-space ratio (less space between them). The text then states that the results of an experiment specifically using the contexts of an inexpensive store (Forever 21) and an upscale one (Tiffany & Co.) suggest that an inexpensive store context may moderate, or lessen, that effect. If Sevilla and

Townsend found that participants in the experiment with Tiffany & Co. and Forever 21 judged the same jewelry items as substantially more valuable when there was lots of space between them than when there was little space between them in the upscale store (Tiffany & Co.) but judged the same jewelry items as only slightly more valuable when there was lots of space between them than when there was little space between them in the inexpensive store (Forever 21), that finding would demonstrate that increased space between products was associated with less of an increase in those products' perceived value at the inexpensive store than at the upscale store. Thus, the finding would support the text's conclusion that a store context associated with inexpensive goods moderates the effect Sevilla and Townsend observed in their earlier experiments.

Choice B is incorrect because if Sevilla and Townsend found that at both upscale and inexpensive stores, participants judged jewelry spaced far apart to be less valuable than jewelry spaced close together, that would show the opposite of the effect the researchers observed in their earlier experiments, not show that an inexpensive store context merely moderates, or lessens, that effect. Choice C is incorrect because this finding wouldn't show that the effect Sevilla and Townsend observed in their initial experiments (that products spaced far apart were perceived as more valuable than the same products spaced close together) is present but lessened in inexpensive retail contexts, as the text suggests. The conclusion in the text rests on determining the difference in jewelry items' perceived value between two spacing conditions within each store and then comparing the difference for the inexpensive store (Forever 21) to the difference for the upscale store (Tiffany & Co.); a finding that compares perceptions of the jewelry items' value between the two stores but for only one type of spacing condition at a time wouldn't provide information about the degree of difference between spacing conditions within each type of store context. Choice D is incorrect because this finding wouldn't show that the effect Sevilla and Townsend observed in their initial experiments (that products spaced far apart were perceived as more valuable than the same products spaced close together) is present but lessened in inexpensive retail contexts, as the text suggests. The conclusion in the text rests on determining the difference in jewelry items' perceived value between two spacing conditions within each store and then comparing the difference for the inexpensive store (Forever 21) to the difference for the upscale store (Tiffany & Co.); a finding that compares perceptions of the relative value of jewelry items between the two stores and for only one type of spacing condition at a time wouldn't provide information about the degree of difference between spacing conditions within each type of store context.

▼ 3.46 — D

Choice D is the best answer because it most effectively uses data from the graph to complete the student's conclusion about banana ripening time with and without ethylene at different temperatures. The graph shows that at 20°C, the gap between the two bars showing ripening times for ethylene-treated bananas and untreated bananas crosses fewer than 2 gridlines (from about 4 days for ethylene-treated bananas to about 5.5 days for untreated bananas). Meanwhile, the graph shows that at 14°C, 16°C, and 18°C, the gap between the bars crosses more than 2 gridlines (from about 8 days for treated bananas to about 11 days for untreated bananas at 14°C; from about 6 days for treated bananas to about 9.5 days for untreated bananas at 16°C; and from about 5.5 days for treated bananas to about 8.5 days for untreated bananas at 18°C). Since the gap between the bars at each of these temperatures crosses more than 2 gridlines, and since each of these gaps is larger than the gap between the bars at 20°C, it can be concluded that ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice A is incorrect because the graph shows that ethylene-treated bananas stored at 20°C ripen more quickly than ethylene-treated bananas stored at the other temperatures do (about 4 days at 20°C vs. about 5.5, 6, and 8 days at 18°C, 16°C, and 14°C, respectively) and that untreated bananas stored at 20°C ripen more quickly than untreated bananas stored at the other temperatures do (about 5.5 days at 20°C vs. about 8.5, 9.5, and 11 days at 18°C, 16°C, and 14°C, respectively). The information in the graph therefore indicates that storing bananas at 20°C speeds up ripening time relative to storing bananas at the other temperatures shown, not that this storage temperature slows ripening time. Choice B is incorrect because the graph shows that as temperature increases, the ripening time of untreated bananas decreases, from about 11 days at 14°C to about 5.5 days at 20°C, with no exceptions to this trend. The graph therefore shows that differences in temperature were associated with absolute differences in ripening time, not that there was no association between differences in temperature and differences in ripening time. Choice C is incorrect because the graph shows that ripening times of ethylene-treated bananas at 14°C and 16°C were about 8 and 6 days, respectively, which is greater than, not less than, ripening times of ethylene-treated bananas at 18°C and 20°C, which were about 5.5 and 4 days, respectively. In other words, bananas treated with ethylene ripen more slowly, not faster, at 14°C and 16°C than at 18°C and 20°C.

▼ 3.47 — D

Choice D is the best answer because it provides the finding that, if true, would most directly support Aburto-Oropeza and colleagues' explanation that species composition differs between Alacranes and Bajos del Norte because of a difference in the regulation of fishing at

the two reefs. The text explains that isolated oceanic reefs are often unprotected and are increasingly attracting commercial and sport fishing activity. According to the text, when Aburto-Oropeza and colleagues surveyed Alacranes, an isolated reef where fishing is prohibited, and Bajos del Norte, an isolated reef that is unprotected, they found that species at the highest trophic level (that is, top predators) make up a smaller proportion of the total mass of all living organisms at Bajos del Norte (10%) than they do at Alacranes (34%). If it were true that commercial and sport fishers tend to disproportionately remove top predators—that is, that they catch them at a higher rate than they catch other species—this would support the explanation that the allowance of fishing at Bajos del Norte and the prohibition of fishing at Alacranes has led to a lower proportion of top predator species remaining in the total biomass at Bajos del Norte.

Choice A is incorrect. To support the researchers' claim that the role of fishing regulations accounts for Alacranes having a greater proportion of species at the highest trophic level (top predators) than Bajos del Norte does, a finding would need to help explain how fishing affects the presence of top predator species in particular; the finding that the two reefs have similar biomass densities (that is, about the same mass of living organisms per unit) and that Alacranes has a much greater total biomass (total mass of living organisms) wouldn't reveal anything about the species composition at each reef or help explain why the *proportion* of top predators in the total biomass at each reef differs. This finding might simply suggest that Alacranes is a larger reef, with more units at that density. Choice B is incorrect because the finding that there are species at the highest trophic level (top predators) at Bajos del Norte that aren't found at Alacranes would reveal only that there are some different species in the reefs' total biomasses (total mass of living organisms) but wouldn't explain why that's the case; this finding wouldn't be clearly connected to the role of fishing regulations, which is what, according to the researchers, accounts for the reefs having different proportions of top predators in their total masses. Choice C is incorrect. To support the researchers' claim that the role of fishing regulations accounts for Alacranes having a greater proportion of species at the highest trophic level (top predators) than Bajos del Norte does, a finding would need to help explain how fishing affects the presence of top predator species in particular, but different costs of reaching the two reefs wouldn't affect fishing behavior at all because fishing isn't even allowed on one of the reefs.

▼ 3.48 — D

Choice D is the best answer because it presents a finding that, if true, would support the researchers' hypothesis that TMAO reduces water's compressibility. The text explains that at great depths in the ocean, extreme pressure compresses the molecular structure of water by

destabilizing the hydrogen bonds between adjacent molecules, thereby allowing water to penetrate proteins and harm the associated organisms. However, deep-sea organisms called piezophiles have adapted to live at these depths and previous studies show a positive correlation between the depth at which a piezophile species lives and the species' level of the compound TMAO. Because this hypothesis links TMAO levels with reduced compressibility of water's tetrahedral molecular structure, a finding that TMAO helps maintain the hydrogen bonds between water molecules under high pressure would strongly support that hypothesis.

Choice A is incorrect. Although the researchers' hypothesis suggests a relationship between TMAO and water molecules' tetrahedral molecular structure, that relationship involves TMAO helping maintain water's tetrahedral molecular structure under high pressure; as presented in the text, the hypothesis doesn't contend that water molecules are impervious to, or incapable of being penetrated by, TMAO. Choice B is incorrect because the text discusses how the molecular structure of water, not TMAO, is compressed under extreme pressure and never addresses how TMAO might be affected by such pressure. Choice C is incorrect because the researchers' hypothesis holds that water under extreme pressure is more resistant, not less, to being compressed when TMAO concentrations are higher. Moreover, the positive correlation mentioned in the text is between TMAO concentrations and the depths at which piezophiles live, not between concentrations of TMAO and the rate at which water's molecular structure compresses as pressure increases.

▼ 3.49 — B

Choice B is the best answer because it presents a finding that, if true, would most directly support Gómez-Bahamón and her team's hypothesis about fork-tailed flycatchers. The text indicates that although two subspecies of the birds live in the same region, the tail feathers of the migrating males make a higher-pitched sound than the tail feathers of the nonmigrating males do. Gómez-Bahamón and her team hypothesize that female fork-tailed flycatchers are attracted to the particular sound made by the tail feathers of males of their own subspecies, which will bring about additional "genetic and anatomical divergence" between the two subspecies. If it were found that the pitch generated by the tail feathers of migrating males is getting higher over successive generations, it would indicate that the shape of the migrating subspecies' tail feathers is diverging further from that of the nonmigrating subspecies. And if females continue to prefer the sounds of the males of their own subspecies, the females of the migrating subspecies will become acclimated to increasingly higher pitches over subsequent generations, causing further divergence between

the subspecies. Thus, if it were found that migrating males' tail feathers were producing higher pitches over time, that would support the researchers' hypothesis.

Choice A is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding is about the shape of wing feathers and how that affects long-distance flight, whereas the hypothesis is about the shape of tail feathers and how that relates to female mate preference. Choice C is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding focuses on how the tail feather sounds communicate different messages, which doesn't address differences between the subspecies or female preferences. Choice D is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. The finding that breeding habits haven't changed for either subspecies does not, by itself, suggest anything about female preferences or divergence between the two subspecies.

▼ 3.50 — C

Choice C is the best answer because it presents a finding that, if true, would most directly support Vikash V. Gayah's claim that eliminating the option to turn left at busy intersections would improve traffic flow and reduce overall travel times. The text begins by describing a problem encountered by drivers in countries with right-hand traffic—namely that drivers wanting to make a left turn must wait for either gaps in oncoming traffic or for designated left-turn signals before proceeding. The resulting backup of vehicles causes increased traffic congestion at busy intersections that slows overall travel times. According to Gayah, eliminating left turns from busy intersections in urban areas would ease the congestion caused by vehicles waiting to turn left. If vehicles spend less time waiting at intersections for left turns, faster overall travel times would result even if some drivers would have to drive slightly longer distances to make the desired left turn. Drivers for package-delivery companies, who presumably spend most of the day driving to destinations across the city in which they are based, would likely provide a good indication of overall traffic patterns across the city. A finding that after a city had eliminated left turns at busy intersections, package-delivery companies were able to complete more daily deliveries on average—which implies faster travel times between package destinations—would therefore support the claim that overall travel times would decrease if left turns were eliminated at busy intersections.

Choice A is incorrect because a finding that a majority of survey respondents agreed with the statement that implementing left-turn signals at all busy intersections made navigating streets in their communities easier wouldn't support the claim that eliminating the option to turn left altogether at some busy intersections—both with and without dedicated left-turn signals—would improve the flow of traffic and overall traffic times. In fact, the text emphasizes that designated left-turn signals contribute to increased congestion because they require drivers to wait to turn left, which results in a backup of vehicles. Installing such signals at all busy intersections would thus potentially compound the problem of congestion, not improve it. Moreover, although making streets easier for drivers to navigate might indicate that left-turn signals are beneficial to drivers, it doesn't indicate that overall travel times would necessarily be reduced. Choice B is incorrect because a study concluding that drivers wait longer to make left turns at intersections without dedicated left-turn signals than at intersections with such signals wouldn't support the claim that eliminating the option to turn left altogether would reduce overall travel times. In fact, the finding would merely support the idea that installing left-turn signals would likely reduce the time drivers spend waiting at busy intersections that didn't previously have left-turn signals, not that overall travel times would be reduced. Choice D is incorrect because the finding that after a restriction eliminating left turns at most intersections took effect, school buses took longer to complete their routes than they did before the restriction was implemented would contradict rather than support the researcher's claim that eliminating left turns would reduce overall travel times.

▼ 3.51 — B

Choice B is the best answer because it describes data from the graph that support Taylor and colleagues' conclusion that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells. The text explains that perovskite solar cells' efficiency at converting light into electricity is diminished by their electron transport layer (ETL), which is applied through spin coating, but that Taylor's team devised a new spray coating method for applying the ETL that improves its power conversion efficiency. The graph displays data on the power conversion efficiency of solar cells in tests conducted by Taylor's team, with bars for both the highest- and lowest-performing ETLs in two data categories: spray coating and spin coating. According to the graph, the lowest-performing ETL applied through spray coating had a power conversion efficiency of between 14% and 16%, while the highest-performing ETL applied through spin coating had a power conversion efficiency of less than 14%. These data confirm that ETLs applied through novel spray coating are more

efficient than those applied though traditional spin coating. Thus, the data support Taylor and colleagues' conclusion about spray coating's potential value.

Choice A is incorrect. Although this claim correctly describes the data in the graph by stating that both the lowest-performing ETL applied through spin coating and the lowest-performing ETL applied through spray coating had a power conversion efficiency greater than 10%, this relationship in the data doesn't support or relate to Taylor and colleagues' conclusion that spray coating promises greater efficiency for solar cells than traditional spin coating does. Choice C is incorrect. This claim does address the greater power conversion efficiency of the highest-performing ETL applied through spray coating, compared with the highest-performing ETL applied through spin coating. However, it also incorrectly cites the value for the efficiency of the highest-performing ETL applied through spray coating as approximately 13%, instead of a value between 14% and 16%, and the value for the efficiency of the highest-performing ETL applied through spin coating as approximately 11%, instead of a value between 12% and 14%, as shown in the graph. Choice D is incorrect because Taylor and colleagues' conclusion is based on the difference in the power conversion efficiency of ETLs applied through spray coating and that of ETLs applied through spin coating, not on the difference between the highest- and lowest-performing ETLs applied through just spray coating.

▼ 3.52 — D

Choice D is the best answer because it provides a quotation that would most directly support the student's claim that the temple's users were aware of how materials used in the temple affected sound and deliberately applied that knowledge to shape the sonic experience of people in the space. The quotation indicates that curtains would, at times, be installed specifically "to minimize reverberation and confine the sound," thereby illustrating that the users of the temple had knowledge of the acoustic properties of materials within the temple (how curtains dampen sound) and employed this knowledge to affect the experience of sound within the space.

Choice A is incorrect. Although the quotation describes a feature of the temple's structure (the high ceiling), it makes no mention of the materials used, the materials' acoustic properties, or temple users' awareness of how materials impact sound quality. Choice B is incorrect because it doesn't address any materials that were used in the temple, let alone their acoustic properties or temple users' awareness of how materials impact sound quality. Choice C is incorrect because it doesn't address any materials that were used in the temple,

let alone their acoustic properties or temple users' awareness of how materials impact sound quality.

▼ 3.53 — C

Choice C is the best answer because it presents a finding that, if true, would support the researchers' hypothesis about the plants' dependence on dissolving rock. The text indicates that the roots of the two plant species grow directly into quartzite rock, where hairs on the roots secrete acids that dissolve the rock. The researchers hypothesize that the plants depend on this process because dissolving rock opens spaces for the roots to grow and releases phosphates that provide the plants with phosphorus, a vital nutrient. If the plants carry out this process of dissolving rock even when the rock already has spaces into which the roots could grow, that would support the researchers' hypothesis because it suggests that the plants are getting some advantage—such as access to phosphorus—from the action of dissolving rock. If the plants don't benefit from dissolving rock, they would be expected to grow in the cracks that already exist, as doing so would mean that the plants don't have to spend energy creating and secreting acids; if, however, the plants create new entry points by dissolving rock even when cracks already exist, that would support the hypothesis that they depend on dissolving rock for some benefit.

Choice A is incorrect because the existence of soil-inhabiting members of the Velloziaceae family with similar root structures to those of the two species discussed in the text wouldn't support the researchers' hypothesis that the species discussed in the text depend on dissolving rock. If other such members exist, that might suggest that the root structures can serve more functions than secreting acids to dissolve rock (since dissolving rock may not be necessary for plants living in soil), but that wouldn't suggest anything about whether the species discussed in the text benefit from dissolving rock. Choice B is incorrect because differences in the proportions of citric and malic acid secreted by the two species would be irrelevant to the hypothesis that the plants depend on dissolving rock. There's no information in the text to suggest that the proportion of each acid has any bearing on the process of dissolving rock or on any benefits the plants might receive from that process. Choice D is incorrect because if the two species thrive on rocks without phosphates, that would weaken the researchers' hypothesis that the plants depend on dissolving rock partly because dissolving rock gives them access to phosphates. If the plants can survive on rocks without getting a vital nutrient by dissolving those rocks, then either the nutrient isn't actually vital for those plants or they can get the nutrient in some way other than by dissolving rocks.

▼ 3.54 — A

Choice A is the best answer because it most effectively uses data from the table to complete the statement. The text explains that mycorrhizal hosts are plants that benefit from the presence of mycorrhizal fungi in the soil and that some such plants produce more mass when grown in the presence of these fungi, while for nonmycorrhizal species the fungi either have no effect or may be harmful. The experiment included two mycorrhizal hosts (corn and marigold) and one nonmycorrhizal species (broccoli). Given the claim in the text that nonmycorrhizal species will see either no difference or a decrease in mass when exposed to mycorrhizal fungi, the student would likely have been surprised by the higher average mass for broccoli grown in the presence of the fungi than the broccoli grown in the soil treated to kill fungi.

Choice B is incorrect. Although this choice accurately describes the corn data from the table, the fact that the mycorrhizal host corn is more massive in the presence of the fungi likely fits with what the student expected and would therefore not be surprising. Choice C is incorrect. Although this choice accurately describes the marigold data from the table, the fact that the mycorrhizal host marigold is more massive in the presence of the fungi is likely what the student expected and thus would not be surprising. Choice D is incorrect because it does not accurately represent the data in the table—when grown in soil treated to kill fungi, corn had an average mass of 3.8 g while broccoli had an average mass of 7g—and because making comparisons among the plants in the no-fungi condition, by itself, does not provide a basis to compare the average mass of mycorrhizal hosts and nonmycorrhizal species grown in the presence of the fungi with those grown in the soil treated to kill fungi.

▼ 3.55 — A

Choice A is the best answer because it most effectively illustrates the claim that Martí argues that a society's spiritual well-being depends on the character of its literary culture. In the quotation, Martí asserts that poetry is "more necessary to a people than industry itself" and that it has the power to provide people with "faith and vigor." He also adds that literature gives people "the desire and strength for life." Therefore, this quotation shows that Martí believes that literature is a societal necessity because it uplifts people and nourishes their spiritual well-being.

Choice B is incorrect. Although this quotation emphasizes the importance of literature, it focuses on how the nature of a society is reflected in that society's literature rather than on literature's value for people's spiritual well-being. Choice C is incorrect. Although this quotation involves an element of spirituality, it doesn't discuss literature. The quotation

instead focuses on humanity's actions. Choice D is incorrect because this quotation mainly focuses on the importance of Walt Whitman rather than on the value of literature in general.

▼ 3.56 — C

Choice C is the best answer because it presents the finding that, if true, would most strongly support Tannen's hypothesis. According to the text, Tannen's hypothesis is that multiple perspectives presented in a noncompetitive format is more informative than a debate between opposing viewpoints is. If participants who saw a panel of three commentators with various views about an issue answered more questions about the issue correctly than did participants who saw a debate, that would support Tannen's hypothesis since it would show that participants who heard multiple varied perspectives were better informed than were participants who heard a debate between opposing viewpoints.

Choice A is incorrect because finding that participants perceived commentators in the debate as more knowledgeable than commentators in the panel is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists. Choice B is incorrect because finding that participants perceived commentators in the panel as more knowledgeable than a single commentator is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists, and Tannen's hypothesis says nothing about how informative single commentators are. Choice D is incorrect because finding that participants who watched a single commentator answered more questions correctly than participants who watched the debate did wouldn't be relevant to Tannen's hypothesis, which is that hearing multiple varying perspectives is more informative than hearing a debate. Tannen's hypothesis says nothing about how informative single commentators are.

▼ 3.57 — A

Choice A is the best answer because it describes data from the table that support the researchers' conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator. The table presents the average time pollinators spent per floral morph. The data in the table shows that for pin flowers, the average time spent per flower by all pollinator genera was higher for flowers that had been

artificially damaged than for intact flowers. By contrast, for thrum flowers, the difference in time spent on intact and damaged flowers is seen for only some pollinator genera. This supports the researchers' conclusion that both floral morph and the genus of the pollinator are factors involved in the effect that floral damage has on time pollinators spend per flower.

Choice B is incorrect. The table shows that pollinators belonging to the genus *Xylocopa* spent less time on both damaged pin flowers and damaged thrum flowers than pollinators belonging to the genus *Osmia* did. Choice C is incorrect. The table shows that the artificial damage to the thrum flowers led to shorter average times spent by pollinators on those flowers for only two of the four pollinator genera represented in the table, not three. Furthermore, this choice doesn't address the effect the artificial damage had on pin flowers; thus, even if accurate, this evidence doesn't support the researchers' conclusion that both floral morph and pollinator genus affect time spent per flower when a flower has been damaged. Choice D is incorrect. While this statement is true based on the information in the table, it doesn't provide evidence that supports the researchers' conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator because this choice provides information only about how much time one pollinator genus spent on intact flowers.

▼ 3.58 — B

Choice B is the best answer. The text explains that firms in the United States often look to municipal governments (local governing bodies) for incentives to expand the firms into those areas. The text then indicates that a team of political scientists hypothesized that those governing bodies are much more likely to respond and offer incentives when the expansion (if there's going to be one) can be announced before an election, since announcing a firm coming to their area can benefit local elected officials. However, the graph—which presents the results of the political scientists' inquiries to municipalities about incentives for a firm in two timing scenarios—shows that the proportion of municipalities that responded to the inquiry or offered incentives didn't substantially differ based on timing. According to the graph, about 200 municipalities responded to the inquiry and about 100 offered an incentive when the announcement could be made before an election (represented by the gray bars) and about 200 municipalities responded to the inquiry and about 100 offered an incentive when the announcement would be made after an election (represented by the black bars). These data weaken the team's hypothesis by demonstrating that municipalities are *not* much more likely to respond and offer incentives when the timing means they could announce an expansion before an election.

Choice A is incorrect. Although the graph does show that a large number of municipalities didn't respond to inquiries in the scenario in which an expansion announcement could be made before an election (represented by the gray bars), the text indicates that the team hypothesized that there would be a substantial difference in municipalities' actions based on timing, and information about only one timing scenario doesn't reveal anything about whether such a difference occurred. Choice C is incorrect. The graph does show that in the scenario in which an expansion announcement could be made before an election (represented by the gray bars), about 200 municipalities responded to inquiries and about 100 offered an incentive, but the text indicates that the team hypothesized that there would be a substantial difference in municipalities' actions based on timing, and information about only one timing scenario doesn't reveal anything about whether such a difference occurred. Choice D is incorrect. Although this choice accurately describes data from the graph for the scenario in which an expansion announcement would be made after an election (represented by the black bars), the text indicates that the team hypothesized that there would be a substantial difference in municipalities' actions based on timing, and information about only one timing scenario doesn't reveal anything about whether such a difference occurred.

▼ 3.59 — A

Choice A is the best answer. This finding would support the conclusion. If stream banks are collapsing into the water at multiple points, then sediment is getting into the water at those various points. This supports the conclusion that the inflow of sediment is very spread out.

Choice B is incorrect. This finding wouldn't support the conclusion. The conclusion is about the influx of sediment being "spatially diffuse," meaning spread out over a large area. The type of sediment wouldn't have an impact on the conclusions. Choice C is incorrect. This finding wouldn't support the conclusion. It doesn't say anything about the influx of sediment being "spatially diffuse" (spread out). Choice D is incorrect. This finding wouldn't support the conclusion. Any sediment that enters downstream of the sampling sites wouldn't end up in the samples, so it wouldn't affect the findings or the conclusion.

▼ 3.60 — A

Choice A is the best answer because it most effectively uses data from the graph to complete the statement about Mahtta et al.'s proposal regarding factors that affect urban land expansion (ULE). According to the text, ULE is influenced by urban population growth and by gross domestic product (GDP) growth per capita. Reasoning that efficient national governments provide urban services and infrastructure needed to attract economic

investment, Mahtta et al. suggest that, as governments become more efficient at providing urban services and infrastructure, GDP growth per capita will account for more ULE and urban population growth will account for less. But according to the graph, Region 1 saw an increase in the percentage attributed to urban population growth from 1970–2000 (between 60 and 65%) to 2000–2014 (between 70 and 75%) and a decrease in the percentage attributed to GDP growth per capita from 1970–2000 (between 35 and 40%) to 2000–2014 (about 25%). Because the percentage attributed to GDP growth per capita decreased (the opposite of what Mahtta et al. claimed would happen if the governments had become more efficient), the data suggest that the governments of Region 1 became less efficient at providing urban services and infrastructure over that period.

Choice B is incorrect. Neither the graph nor the text gives the regions' relative levels of economic growth or what effect Mahtta et al. would expect such growth to have. Furthermore, Mahtta et al.'s proposal suggests that Region 1's decline in the percentage of ULE attributed to GDP growth per capita from 1970–2000 (between 35 and 40%) to 2000–2014 (about 25%) would suggest decreasing, not increasing, government efficiency over this time. Choice C is incorrect. Neither the text nor the graph provides information about the relative efficiencies of different governments in Region 2. Choice D is incorrect. Mahtta et al.'s proposal suggests that more efficient governments will have a higher percentage of their ULE driven by GDP growth per capita and a lower percentage driven by urban population growth. For Region 2, the percentage of ULE attributed to GDP growth per capita increased from 1970–2000 (between 10 and 15%) to 2000–2014 (between 45 and 50%), but the opposite is true for Region 1, which saw the percentage of ULE attributed to GDP growth per capita decline over the same period. Thus, whereas the data suggest governments in Region 2 became more efficient, the data for Region 1 suggest that those governments became less efficient, not more.

▼ 3.61 — C

Choice C is the best answer. Electra states that she "knows her own passion," which shows that she's aware of the intensity of her grief. But she also claims that the "causes are so dire"—meaning the reasons for her grief are so awful—that she can't let it go, which shows that she believes her grief is justified.

Choice A is incorrect. This quotation doesn't show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn't suggest that she believes she has a legitimate reason for feeling that way. Choice B is incorrect. This quotation doesn't show that Electra believes her grief is justified. It shows that Electra is aware of its intensity,

but it doesn't suggest that she believes she has a legitimate reason for feeling that way. Choice D is incorrect. This quotation doesn't show that Electra believes her grief is justified. It shows that Electra is aware of the intensity of her grief, but it doesn't suggest that she has a legitimate reason for feeling that way.

▼ 3.62 — D

Choice D is the best answer. The text tells us that the farther the correlation is from 0, the "stronger" it is (doesn't matter if it's negative or positive). The table shows that the correlations with velocity are farther from zero than the correlations with density, which supports the conclusion that the correlations with velocity are stronger.

Choice A is incorrect. This choice doesn't support the conclusion. It makes an "apples to oranges" comparison by comparing density and velocity correlations across features instead of comparing them for each feature. Choice B is incorrect. This choice doesn't support the conclusion. It doesn't include the density correlations for comparison. Choice C is incorrect. This choice doesn't support the conclusion. The text tells us that the farther the correlation is from 0, the "stronger" it is: it doesn't matter for "strength" whether it's negative or positive.

▼ 3.63 — D

Choice D is the best answer because it presents a finding that, if true, would support Shcherbakova and colleagues' claim that the outcome of their study is inconsistent with the linguistic niche hypothesis (LNH). The text explains that the LNH holds that there is an inverse relationship between the prevalence of non-native speakers of a language (exotericity) and the grammatical complexity of that language—that is, that as the number of non-native speakers increases, grammatical complexity decreases, and vice versa. According to the text, Shcherbakova and colleagues focused on two positive indications of grammatical complexity—fusion and informativity—and analyzed their occurrence in over 1,300 languages. If the researchers found a slightly positive correlation between fusion and exotericity and between informativity and exotericity—meaning that to some extent, grammatical complexity increases as the number of non-native speakers of a language increases—their outcome would not be consistent with the assumption that exotericity and grammatical complexity are inversely related (the LNH).

Choice A is incorrect because it wouldn't be possible to say that a finding of a slightly negative correlation between grammatical complexity and both fusion and informativity is inconsistent or consistent with the LNH, since the finding would address only grammatical

complexity (given that fusion and informativity are aspects of grammatical complexity) and wouldn't move beyond that factor to address its relationship to the prevalence of non-native speakers of a language (exotericity), which is the relationship the LNH focuses on. Choice B is incorrect because a finding of a slightly negative correlation between grammatical complexity and the prevalence of non-native speakers of a language (exotericity)—meaning that as the number of non-native speakers increases, grammatical complexity somewhat decreases, and vice versa—would be consistent, not inconsistent, with the LNH, since the text indicates that according to the LNH, there is an inverse relationship between grammatical complexity and exotericity; a negative correlation reflects an inverse relationship. Choice C is incorrect because it wouldn't be possible to say that a finding of a slightly positive correlation between grammatical complexity and fusion is inconsistent or consistent with the LNH, since the finding would address only grammatical complexity (given that fusion is a positive indication of grammatical complexity) and wouldn't move beyond that factor to address its relationship to the prevalence of non-native speakers of a language (exotericity), which is the relationship the LNH focuses on.

▼ 3.64 — D

Choice D is the best answer because it presents a finding that, if true, would support Asmerom and colleagues' conclusion that the ITCZ may have expanded northward and southward rather than shifting south during the Little Ice Age. The text indicates that the ITCZ, a band of clouds in the tropics that is a significant rainfall source, can change position. Data from Peru's Huagapo Cave suggest that the ITCZ shifted south during the Little Ice Age. But according to the text, if the ITCZ moved into South America in that way, then Central America should have been drier than climate models suggest it was. In other words, rainfall should have been reduced in Central America because the ITCZ, a significant rainfall source, had shifted into South America, but climate models do not show such a reduction in Central America. The text goes on to say that Asmerom and colleagues tried to resolve this apparent conflict by collecting data from Yok Balum cave in Central America and comparing them with data from Huagapo, which led the researchers to conclude that the ITCZ may have expanded both northward and southward rather than simply shifting south. If it is true that Yok Balum in Central America and Huagapo in South America show strongly correlated patterns of high rainfall during the Little Ice Age, such a finding would support Asmerom and colleagues' conclusion by suggesting that the two areas were affected by the same rainfall source, and thus that the ITCZ may have expanded rather than shifted.

Choice A is incorrect because there is no information in the text about how, if at all, the ITCZ affects temperature in areas where it is located. Rather, the text states that temperature

variations across Earth's hemispheres can shift the position of the ITCZ. Finding that neither Yok Balum nor Huagapo data show evidence of significant local variations in temperature during the Little Ice Age would have no clear bearing on Asmerom and colleagues' claim. Choice B is incorrect because finding that both Yok Balum and Huagapo experienced prolonged dry conditions during the Little Ice Age would not support Asmerom and colleagues' conclusion that the ITCZ, a major source of rainfall, may have expanded northward and southward rather than simply shifting south. Dry conditions in both locations would suggest that the ITCZ did not cover either location. Additionally, finding that temperatures were elevated in both locations would have no clear bearing on Asmerom and colleagues' conclusion, since there is no information in the text that indicates how, if at all, the ITCZ affects temperature. Choice C is incorrect because finding that Yok Balum experienced prolonged dry conditions at the same time that Huagapo experienced high rainfall would weaken Asmerom and colleagues' conclusion, not strengthen it. Such a finding would suggest that the ITCZ shifted south and left Central America dry rather than expanding both northward and southward.

▼ 3.65 — A

Choice A is the best answer because it uses data from the table to give the range of velocities for the LMC from the 1980 value (344 km/s) to the 2006 value (378 km/s), thereby effectively completing the text. The text indicates that before 2006, all the estimated velocities of the LMC were within the range necessary to maintain orbit around the Milky Way galaxy. It then indicates that, according to Besla and colleagues, the 2006 estimate of 378 km/s is too high to maintain that orbit. This strongly implies that if the 1980 value (344 km/s) is below the orbital threshold, and if Besla and colleagues are correct that the 2006 value (378 km/s) is above that threshold, the maximum orbital velocity for the LMC must be somewhere in the range from above 344 km/s to below 378 km/s.

Choice B is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first estimate that exceeded the velocity needed to maintain orbit around the Milky Way. Thus, the 1980 estimate of 344 km/s and the 1994 estimate of 297 km/s must both be below the maximum possible orbital velocity for the LMC. Choice C is incorrect. The text states that Besla and colleagues' analysis found that the velocity from the 2006 study (378 km/s) was too high for the LMC to maintain orbit around the Milky Way. Therefore, if a velocity of 378 km/s is too high, an even higher velocity will also be too high. Choice D is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first to exceed the velocity range required to maintain orbit around the Milky Way. Thus, the

1994 estimate of 297 km/s must be below the maximum possible orbital velocity for the LMC.

▼ 3.66 — D

Choice D is the best answer. The researchers concluded that as we approach maximum plausible surface temperatures, there will be a larger percentage of sites where thermal pollution could contribute to meeting home heating needs. By showing that only a small percentage of homes can currently use thermal pollution for home heating, and that this percentage would grow much larger at maximum plausible surface temperatures, this choice supports the researchers' conclusion.

Choice A is incorrect. We do not know how many sites could have all (i.e., 100%) of their local heating needs met by thermal pollution, as the graph only classifies sites by whether "0%," "Up to 25%," and "More than 25%" of heating needs could be met. Choice B is incorrect. The graph is not depicting need for supplemental heating from thermal pollution, but rather potential to use thermal pollution for supplemental heating. Choice C is incorrect. The graph indicates that, at current surface temperatures, less than 10% of sites can meet 25% of local home heating needs and that more than 80% of sites cannot meet any local home heating needs.

▼ 3.67 — D

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers' hypothesis about the connection between the dusky shark population decline and the eastern oyster population decline. The text indicates that although dusky sharks don't usually eat eastern oysters, they do consume cownose rays, which are the main predators of eastern oysters. An increase in the abundance of cownose rays in the region in response to a decline in the abundance of dusky sharks would directly support the researchers' hypothesis: a higher number of cownose rays would consume more eastern oysters, driving down the oyster population.

Choice A is incorrect because a finding that there's an association between a decline in the regional abundance of some of dusky sharks' prey and the regional abundance of dusky sharks wouldn't directly support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. Although such a finding might help explain why shark abundance has declined, it would reveal nothing about whether the shark decline is related to the oyster decline. Choice B is incorrect because a finding that eastern oyster abundance tends to be greater when dusky sharks and cownose rays are present

than when only dusky sharks are present wouldn't support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. The text indicates that the sharks prey on the rays, which are the main predators of the oysters; if oyster abundance is found to be greater when rays are present than when rays are absent, that would suggest that rays aren't keeping oyster abundance down, and thus that a decline in rays' predators, which would be expected to lead to an increase in the abundance of rays, wouldn't bring about a decline in oyster abundance as the researchers hypothesize. Choice C is incorrect because a finding that consumption of eastern oysters by cownose rays increased substantially before dusky sharks declined in regional abundance wouldn't support the researchers' hypothesis that the decline in dusky sharks has led to a decline in eastern oysters in the region. Such a finding would suggest that some factor other than shark abundance led to an increase in rays' consumption of oysters and thus to a decrease in oyster abundance, thereby weakening the researchers' hypothesis.

▼ 3.68 — D

Choice D is the best answer because it most effectively uses a quotation from *O Pioneers!* to illustrate the claim that Alexandra Bergson takes comfort in understanding the world around her. In the quotation, Alexandra is described as enjoying looking at the stars and feeling a "sense of personal security" when she contemplates nature's order and its governing laws. This suggests that Alexandra takes comfort in understanding the world around her.

Choice A is incorrect because the quotation expresses how Alexandra Bergson attempts to meet difficult situations with determination, not how she takes comfort in understanding the world around her. Choice B is incorrect because the quotation expresses "how much the country meant to" Alexandra Bergson, not how she takes comfort in understanding the world around her. In detailing some of the wildlife surrounding Alexandra, the quotation conveys that nature is important to her but not necessarily that it gives her comfort. Choice C is incorrect because the quotation describes Alexandra driving her wagon down a highway at night; it doesn't describe how she takes comfort in understanding the world around her or address how she's feeling as she drives off.

▼ 3.69 — B

Choice B is the best answer. By showing that "any difference" in taste was quickly overcome by "the force of [Marianne's] arguments," this choice effectively demonstrates Marianne's "ability to persuade others."

Choice A is incorrect. This choice doesn't effectively illustrate the claim. This choice shows that Marianne and John share an interest in music and dancing, but it doesn't provide evidence of Marianne's "ability to persuade others." Choice C is incorrect. This choice doesn't effectively illustrate the claim. This choice shows that Marianne enjoys talking about her interests, but it doesn't provide evidence of Marianne's "ability to persuade others." Choice D is incorrect. This choice doesn't effectively illustrate the claim. This choice shows that Marianne and John share many interests and generally agree on music and dancing, but it doesn't provide evidence of Marianne's "ability to persuade others."

▼ 3.70 — C

Choice C is the best answer because it most directly supports the text's claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. The text challenges the characterization of Stevens's preserved sandpaintings as mere "static objects" that cannot "authentically represent" a practice defined in part by the absence of an end product; it claims instead that this "overly object-focused" interpretation overlooks how Stevens preserved the essential characteristics of traditional *ikaah* practice. This quotation supports the claim by explaining that Stevens's sandpaintings should be understood "not as self-contained objects but as reminders of the public creations of the sandpaintings," during which Stevens "conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process." The fact that Stevens created the sandpaintings in a public setting and invited viewers to attentively engage with this process supports the claim that he sought to portray *ikaah* as a dynamic practice rather than as a static product. Further, the inherently impermanent nature of such a performance evokes the ephemerality that characterizes *ikaah*.

Choice A is incorrect because it contradicts rather than supports the text's claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. This quotation argues that Stevens's preserved sandpaintings "should not be confused with authentic *ikaah*" and emphasizes that authentic *ikaah* "cannot be extricated from a practice that is intentionally and necessarily transitory." This view aligns with the "overly object-focused" perspective that the text argues against, rather than supporting the claim that Stevens successfully conveyed *ikaah*'s ephemeral and dynamic characteristics. Choice B is incorrect because it focuses on reconciling the tension between ephemeral and permanent forms rather than supporting the claim that Stevens actively conveyed *ikaah*'s ephemeral and dynamic characteristics. While this quotation acknowledges an apparent contradiction, it suggests that Stevens was accommodating US preferences for permanent

artworks; it doesn't demonstrate how he communicated the ephemeral and dynamic nature of traditional *ikaah* practice. Choice D is incorrect. Although the quotation mentions Stevens's "temporary and performative role" as a gift-giver, it doesn't address how Stevens conveyed the ephemeral and dynamic characteristics of *ikaah* itself during his ambassadorial activities. The quotation focuses on Stevens's role in transmitting knowledge and representing cultural dynamism but doesn't specifically support the claim about his efforts to communicate the essential characteristics of *ikaah* practice. Further, the quotation incorrectly implies that Stevens shared his knowledge of preservative additives; the text doesn't establish that he revealed how the sandpaintings were preserved.

▼ 3.71 — C

Choice C is the best answer because it most logically completes the text based on supporting data in the graph. The text indicates that in the fly *D. melanogaster*, *DptA* and *DptB* are genes that encode peptides that both fight pathogens and promote beneficial microbes. Researchers tested *D. melanogaster*'s resistance to *P. rettgeri* and *A. sicerae* bacteria based on which variation of the peptide-encoding gene the flies exhibit: *DptA* silenced (referred to as type A), *DptB* silenced (type B), or both silenced (type AB). The text also indicates that resistance to *P. rettgeri* correlates with *DptA* activity but not with *DptB* activity (which would manifest as type B flies surviving at a higher rate than other fly types when exposed to *P. rettgeri*). The graph shows the post–*A. sicerae* infection results, which indicate that *DptB* activity was most strongly associated with survival, whereas *DptA* activity was not (manifesting in the graph as the type A flies having greater survival rates than the other fly types). In other words, when *DptA* activity was silenced, the flies showed relatively high survival rates, but when *DptB* activity was silenced, whether on its own or in conjunction with *DptA* activity being silenced, survival rates were low, suggesting that *DptB* may have developed as a specific defense against *A. sicerae*.

Choice A is incorrect. The graph suggests that *DptA* activity is associated with a low rate of survival, not a high one. Furthermore, the graph shows results for flies where *DptA* alone was silenced, *DptB* alone was silenced, and both were silenced and thus does not show any flies with activity in both *DptA* and *DptB*, which would be necessary to determine whether *DptA* conferred defense against *A. sicerae* in the presence of *DptB*. Choice B is incorrect. Only two bacteria species were considered in the text: *P. rettgeri* and *A. sicerae*. The text and graph taken together suggest that activity in *DptA* is associated with resistance to *P. rettgeri* while *DptB* activity is not, and that *DptB* activity is associated with resistance to *A. sicerae* while *DptA* is not. There is no further information to suggest one genetic type confers resistance to a greater number of pathogens than the other. Choice D is incorrect. The graph

does not address flies with activity in both *DptA* and *DptB*. All flies represented in the graph had one or both of *DptA* and *DptB* silenced, or inactive.

▼ 3.72 — D

Choice D is the best answer because it describes data from the table that support the underlined conclusion. The text accompanying the table states that delta 15-N values (a measure of the ratio of two nitrogen isotopes) can be used as an indirect means of measuring the ratio of human wastewater to water from uncontaminated sources—the higher the delta 15-N values, the greater the presence of human wastewater. The text goes on to mention an experiment conducted by Camacho-Cruz and colleagues measuring delta 15-N values in seagrass on the Yucatan peninsula at several sites, four of which are presented in the table accompanying the text. Two of the sites noted in the table, Akumal Bay and Tulum, had intermediate tourism development, and two others, Mahahual and Xahuayxol, had low tourism development. It is reasonable to infer that sites with intermediate tourism have a greater human presence than those with low tourism and therefore are subject to greater amounts of wastewater and exhibit higher delta 15-N values in seagrass. However, Camacho-Cruz and colleagues found that the increase in delta 15-N values was not constant. The table supports this conclusion because the two intermediate-tourism sites (Akumal Bay and Tulum) had delta 15-N values that fluctuated considerably—both had higher delta 15-N values than the two low-tourism sites (Mahahual and Xahuayxol) in October 2016 and October 2017, but in February 2017 the delta 15-N values for the intermediate-tourism sites dropped significantly (in fact, Mahahual, a low-tourism site, had a higher delta 15-N value for that period). The fluctuations in values in Akumal Bay and Tulum support the conclusion that these sites experienced influxes of human wastewater intermittently.

Choice A is incorrect because it inaccurately describes data from the table: in February 2017 the delta 15-N value in Akumal Bay was higher than the delta 15-N value in Xahuayxol, not lower. Choice B is incorrect because the absence of data for delta 15-N levels in Akumal Bay in February 2016 would neither prove nor disprove that the presence of human wastewater at Akumal Bay and Tulum was not constant. Despite the unavailable data, there is still enough information to conclude that delta 15-N levels at both sites fluctuated significantly and that influxes of human wastewater therefore happened intermittently. Choice C is incorrect because a constancy in delta 15-N values would not indicate that the influx of human wastewater was intermittent, but rather the opposite. Moreover, the two periods selected are not indicative of general trends in the whole table (for example, the values

fluctuated significantly in Akumal Bay between February 2017 and October 2017, and in Tulum between October 2016 and February 2017).

▼ 3.73 — B

Choice B is the best answer. The claim is that low anger does not negatively affect readers' perceptions of the review, while high anger does negatively affect readers' perceptions of the review. This choice accurately expresses the supporting data from the "helpfulness of review" part of the graph: that low-anger reviews were rated as slightly more helpful than no-anger reviews, while high-anger reviews were rated as less helpful than no-anger reviews.

Choice A is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' ratings of the reviewed product are not relevant. Choice C is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' attitude towards the reviewed product is not relevant. Choice D is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' attitude towards the reviewed product is not relevant.

▼ 3.74 — A

Choice A is the best answer because it best describes data from the graph that support the researchers' conclusion about whitefly-induced defenses in tomato plants. According to the text, tomato plants produce airborne chemicals that prompt nearby tomato plants to increase their production of appropriate defensive chemicals, such as jasmonic acid in response to herbivory and salicylic acid in response to microbial infection. The text goes on to indicate that researchers concluded that whiteflies induce tomato plants to emit chemicals that cause nearby plants to defend against microbial infection instead of herbivory from whiteflies. The graph shows that control tomato plants (plants that weren't exposed to any chemicals from nearby tomato plants and weren't subsequently infested with whiteflies) produced about 140 nanograms of jasmonic acid per gram of dry weight and about 120 nanograms of salicylic acid per gram of dry weight, which indicates the baseline levels of these defensive chemicals in tomato plants. The graph also shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when subsequently infested by whiteflies, which is more than the amount produced by the control plants. This suggests that when infested with whiteflies, tomato plants that weren't previously exposed to air from nearby whitefly-infested plants begin producing higher levels of jasmonic acid to defend against the infestation. Finally, the graph also shows that

plants exposed to air from whitefly-infested plants and then subsequently infested by whiteflies produced approximately 100 nanograms of jasmonic acid per gram of dry weight (which is less than control plants did) and approximately 175 nanograms of salicylic acid per gram of dry weight (which is more than control plants did). This suggests that whiteflies cause tomato plants to emit chemical signals that decrease nearby plants' defenses against whitefly herbivory relative to baseline levels and redirect their defenses toward microbial infection, thereby supporting the researchers' conclusion that whiteflies cause tomato plants to emit airborne chemical signals that induce nearby plants to preferentially defend against microbes even when experiencing herbivory from whiteflies.

Choice B is incorrect. Although the graph shows that tomato plants exposed to whitefly-infested plants produced less jasmonic acid than salicylic acid (about 100 nanograms of jasmonic acid per gram of dry weight vs. about 175 nanograms of salicylic acid per gram of dry weight), and although this might suggest that those plants are therefore better defended against microbial attack than against herbivory, this wouldn't be sufficient to conclude that this is the case: without reference to the relative levels of defensive chemicals produced by the plants in the other experimental conditions, it would be impossible to conclude that exposure to air from whitefly-infested plants either induces or diminishes these plants' defenses to different kinds of threats. Furthermore, the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight and about 130 nanograms of salicylic acid per gram of dry weight, not that they produced the same amount of the two acids. Choice C is incorrect because it inaccurately describes the data in the graph. The graph shows that plants in all three conditions produced both jasmonic and salicylic acid, not that plants exposed to air from whitefly-infested plants produce no jasmonic acid at all. Choice D is incorrect. Although the graph shows that plants exposed to air from whitefly-infested plants produced about 100 nanograms of jasmonic acid per gram of dry weight when infested, which is less than the amount produced by control plants (about 140 nanograms per gram of dry weight), and although the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when infested (which is more than the amount produced by control plants) and about 130 nanograms of salicylic acid per gram of dry weight (which is more than the approximately 120 nanograms per gram of dry weight produced by control plants), this wouldn't support the researchers' conclusion that plants exposed to air from whitefly-infested plants preferentially defend against microbial infection—that is, that they increase their levels of salicylic acid and decrease their levels of jasmonic acid relative to control plants. Rather, the data would suggest that exposure to air from whitefly-infested plants reduces plants' defenses against herbivory, but these data

wouldn't indicate that whiteflies induce plants to redirect their defenses to guard against microbial infection. Furthermore, the data concerning plants exposed to air from whitefly-free plants merely indicate that these plants increase their defenses against both herbivory and microbial infection when infested by whiteflies.

▼ 3.75 — D

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers' conclusion that the transition from a stagnant lid regime to a tectonic plate regime occurred around 3.2 billion years ago. The text explains that early in Earth's history, Earth exhibited a stagnant lid regime in which there's no interaction between the lithosphere and the underlying mantle. The text further explains that, by contrast, once Earth began to exhibit a tectonic plate regime, its lithospheric and mantle material began to mix. If mantle-derived rocks younger than 3.2 billion years contain material not found in older mantle-derived rocks, that material must have originated somewhere other than the mantle. And if this material is found in both older and contemporaneous lithospheric rocks, that would imply that the lithosphere was able to mix with mantle material beginning around 3.2 billion years ago, as the researchers concluded.

Choice A is incorrect. The text gives no basis for comparing the quantities of lithospheric and mantle-derived rocks. Choice B is incorrect. The text gives no basis for comparing the material makeup of lithospheric rocks to that of mantle-derived rocks. Choice C is incorrect. A positive correlation between the age of lithospheric rocks and these rocks' chemical similarity to mantle-derived rocks would mean that the oldest rocks would be the most similar, which contradicts the text's claim that lithospheric and mantle-derived rocks were completely separate until 3.2 billion years ago. If the researchers' conclusion about the onset of tectonics on Earth is correct, then younger lithospheric rocks would show greater chemical similarity to mantle-derived rocks than older lithospheric rocks do.

▼ 3.76 — C

Choice C is the best answer because it accurately describes data from the table that support Barrett and Rayfield's suggestion about bite force estimates. According to the text, Barrett and Rayfield believe that estimates of dinosaur bite force may be strongly influenced by the methods used to produce them—that is, that different methods may produce significantly different results. The table shows that the studies by Bates and Falkingham and by Cost et al. used the same estimation method (muscular and skeletal modeling) and produced similar bite force estimates (approximately 35,000–57,000 newtons and 35,000–63,000

newtons, respectively). The study by Meers, however, used body-mass scaling and produced a much higher bite force estimate (183,000–235,000 newtons), while the study by Gignac and Erickson used tooth-bone interaction analysis and produced a much lower bite force estimate (8,000–34,000 newtons). The fact that one method produced similar estimates in two different studies and that two different methods used in other studies produced substantially different estimates supports the idea that dinosaur bite force estimates are significantly influenced by the methodology used to produce them.

Choice A is incorrect because it inaccurately describes data from the table. The table does show that the studies by Meers and by Cost et al. used different estimation methods and produced very different ranges of estimated dinosaur bite force, which would support Barrett and Rayfield's suggestion that different methodologies may produce significantly different estimates. However, the table doesn't show that the study by Meers produced the lowest estimated maximum bite force while the study by Cost et al. produced the highest. In fact, the study by Meers estimated a maximum bite force of approximately 235,000 newtons, which is the highest of all the estimated maximums. Choice B is incorrect. Although the data from Gignac and Ericson's study are accurately described, a single set of findings from one study using only one methodology can't show that different methodologies may produce significantly different dinosaur bite force estimates, as Barrett and Rayfield suggest. Choice D is incorrect. Although the table shows that the maximum bite force estimated by Cost et al. was higher than that estimated by Bates and Falkingham, the difference is relatively small; in fact, both teams estimated a minimum bite force of approximately 35,000 newtons and a maximum bite force close to approximately 60,000 newtons. Because these findings demonstrate that a single methodology (muscular and skeletal modeling) produced similar overall results in two studies, the findings don't support Barrett and Rayfield's suggestion that different methodologies may produce significantly different dinosaur bite force estimates.

▼ 3.77 — B

Choice B is the best answer because it describes data from the graph that best support the student's assertion that initial efforts at trade liberalization in China were shaped by firms having limited capital (assets available for use) and that this situation resolved during the 2000s. The text explains that an approach to trade liberalization involves engaging in processing imports, one type of which doesn't require payment to a trade partner (processing with assembly) and one type of which requires upfront payment to a trade partner for raw materials (processing with inputs). The graph, which presents China's imports for ordinary imports and both types of processing imports in the years 2000, 2003, and

2006, shows that while processing imports with assembly rose from about 250 hundred million dollars in 2000 to about 750 hundred million dollars in 2006, processing imports with inputs rose much more sharply, increasing from approximately 650 hundred million dollars in 2000 to about 2,300 hundred million dollars in 2006. Because processing with inputs requires firms to pay for materials (expending capital) and processing with assembly doesn't, the sharper rise in processing imports with inputs suggests that Chinese firms' assets—and thus their ability to engage in that type of processing imports—were relatively limited in (and before) 2000 and then substantially increased from 2000 to 2006. In other words, the data suggest that the situation of having limited capital resolved during the 2000s.

Choice A is incorrect because the graph indicates that ordinary imports were greater than both types of processing imports in 2006, not that processing imports with inputs were greater than ordinary imports and processing imports with assembly that year. Choice C is incorrect because the observation that ordinary imports were greater than both types of processing imports in 2000, 2003, and 2006 doesn't address a change within any type of imports from 2000 to 2006, and an indication of a change in that period that might be related to the availability of assets is needed to support the assertion that the situation of having limited capital resolved during the 2000s. Choice D is incorrect because the fact that processing imports with assembly were greater at the end of the period from 2000 to 2006 than processing imports with inputs were at the start of the same period doesn't address a change within either type of imports during the period, and an indication of such a change that might be related to the availability of assets is needed to support the assertion that the situation of having limited capital resolved during the 2000s.

▼ 3.78 — B

Choice B is the best answer. In this quotation, Dr. Proudie is described as "fondly flatter[ing] himself" that he has gifts that "must be sure to recommend him to a yet higher notice." In other words, he expects his skills to push him to greater fame and success. This implies an exaggerated sense of his own abilities, which matches the claim we're trying to support.

Choice A is incorrect. This quotation doesn't describe Proudie's view of himself, nor does it paint him in an especially flattering light. Instead, by saying his mental powers and business skill are not "proved," it implies that he is actually dim-witted and bad at business. Choice C is incorrect. This choice describes Proudie's closeness to power and importance, but it doesn't show what Proudie thinks of himself. Proudie is not describing himself or his abilities here. The narrator is. Choice D is incorrect. While this quotation offers Proudie very mild

praise, it doesn't show what Proudie thinks of himself or his own abilities, which is what the claim focuses on.

▼ 3.79 — A

Choice A is the best answer. While many networks can perform navigation tasks, or even mimic grid cells, it doesn't mean they're actually behaving like biological brains—this finding suggests that the rules that govern neural network behavior are completely unlike the way real brains work.

Choice B is incorrect. Although it mentions the rules that are programmed into the networks, this finding wouldn't clarify whether or not these rules have anything to do with the function of biological brains. Choice C is incorrect. This choice suggests that neural networks are modeled after multiple types of brain cells, which sidesteps the question of whether these rule-based networks are genuinely similar to biological brains. Choice D is incorrect. This choice doesn't address the key point of the claim, which is that the apparent similarity between neural networks and biological brains is only due to the rules programmed into the networks. It focuses on training tasks, not the originally programmed rules.

▼ 3.80 — D

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it's advantageous for plants at an early stage of development to grow in patches of vegetation.

Choice A is incorrect because the statement that less than 75% of juvenile plants were found growing in patches of vegetation, while true, doesn't clearly support the claim that the plants growing in close proximity to other plants gained an advantage at an early

developmental stage. Saying that less than 75% of plants were found in patches doesn't indicate how the percentage growing in patches compares with the percentage that would be expected to grow in patches on the basis of chance alone, which is the information necessary to evaluate whether the claim in the text has support in the table. Put another way, if the percentage of plants found growing in patches was 15% or less, it would be true that less than 75% were found in patches, but the data would in fact weaken the claim in the text, not strengthen it, since the data would show that growing in patches wasn't advantageous. Choice B is incorrect because only 12 plants of this species were found growing in patches, which was the lowest number of any species, not the greatest number. Additionally, even if it were true that this species had the greatest number of plants growing in patches, the finding would be irrelevant to the claim that plants of all five species gained an advantage by growing in close proximity to other plants. Choice C is incorrect because 59.1% of the plants of these species were found growing in patches, which is a far greater percentage, not a lower percentage, than what would be expected if plants were randomly distributed (around 15%). Additionally, if it were true that the percentage of plants growing in patches was lower for these species than what would be expected from chance alone, that finding would weaken, not strengthen, the claim that growing in patches is advantageous.

▼ 3.81 — D

Choice D is the best answer because it describes data in the graph that support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. According to the text, payload capacity is calculated by using a ratio of a robot's holding force to the robot's weight, and higher ratios indicate a greater payload capacity. The Ultra-Fast Robot Hand has a holding force of 56 newtons, four times greater than that of the Yale Model T. Additionally, the graph shows that the Ultra-Fast Robot Hand has a weight of approximately 500 grams, slightly more than the Yale Model T's weight of approximately 480 grams. Therefore, the Ultra-Fast Robot Hand has a higher ratio of holding force to weight than the Yale Model T. Since higher ratios correspond to greater payload capacity, the information from the graph indicating that the Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T combined with the information in the text ultimately supports the conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Choice A is incorrect. Although, according to the graph, it's true that both the Ultra-Fast Robot Hand and the Yale Model T weigh more than 450 grams, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a

distinction, between the two robots. Choice B is incorrect. Although, according to the graph, it's true that the Ultra-Fast Robot Hand and the Yale Model T both weigh more than the Permanent Magnet Hand does, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the Ultra-Fast Robot Hand and the Yale Model T. Furthermore, the comparison to the Permanent Magnet Hand is irrelevant to the claim about the relative ratios and payload capacities of the Ultra-Fast Robot Hand and the Yale Model T. Choice C is incorrect. Although the text states that the Yale Model T has a lower holding force than the Permanent Magnet Hand, the graph provides no information about holding force. Moreover, information about the Permanent Magnet Hand is irrelevant to the conclusion by Meng and colleagues, which only concerns the Ultra-Fast Robot Hand and the Yale Model T.

▼ 3.82 — D

Choice D is the best answer because it presents a finding that, if true, would support the researchers' prediction about the language nest model of education. The text states that Morcom and Roy studied the effects of the language nest model of education on students at an Anishinaabe school, and they found that the model—which is used with students during pre-K or elementary school—increased students' fluency in the Anishinaabe language and pride in Anishinaabe culture. The researchers predicted that the students' positive early experiences with the Anishinaabe language would lead them to be more likely to later share the language with younger generations. If former students maintain full fluency and cultural pride after finishing secondary and higher education, it follows that they would be both able and motivated to share what they know with others; this would likely result in a higher probability of transmitting the language to younger generations, as the researchers predict.

Choice A is incorrect because finding that Anishinaabe adults who didn't attend the school feel approximately the same degree of cultural pride as those adults who did attend wouldn't support the researchers' prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people. Choice B is incorrect because finding that new students experience increased performance in language fluency and academics would suggest that the school has a positive effect on students when they attended but wouldn't reveal anything about those students' later actions as adults (such as their likelihood of sharing their knowledge with younger generations). Choice C is incorrect because finding that Anishinaabe adults who attended the school are equally likely to stay in the community as

adults who didn't attend the school wouldn't support the researchers' prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people.

▼ 3.83 — A

Choice A is the best answer because it most effectively uses data from the graph to support the researchers' recommendation that Sururu shell aggregate (SSA) substitution be done in relatively small proportions to avoid compromising the compressive strength of mortar mixtures. The text explains that a research team investigated whether SSA can serve as a substitute for sand in mortar mixtures. The graph presents the compressive strength (in megapascals) of mortar mixtures with SSA replacing 0% (control), 10%, 20%, and 40% of the sand on day 7 and day 28 of curing. The graph shows that the compressive strength of the 10% SSA mixture was approximately 25 megapascals on day 7, which is greater than the control mixture's approximately 22 megapascals on the same day. The graph also shows that the 10% SSA mixture's compressive strength was slightly more than 25 megapascals on day 28, which is only slightly less than the control's compressive strength on the same day (about 26 megapascals). Therefore, the graph shows that the compressive strength of the 10% SSA mixture was greater than that of the control on day 7 and roughly equal to that of the control on day 28, suggesting that substituting only 10% of the sand with SSA avoids compromising the mortar's compressive strength. The graph also shows that on day 7, the 20% and 40% SSA mixtures had compressive strengths of about 19 and 13 megapascals, respectively (both lower than the control's), and on day 28, they had compressive strengths of about 22 and 15 megapascals, respectively (once again lower than the control's). Thus, the data support the recommendation by indicating that substituting relatively small proportions of sand with SSA in mortar mixtures doesn't compromise strength compared with the control, whereas substituting larger proportions does.

Choice B is incorrect because the comparison it makes isn't relevant to the researchers' recommendation. Although the graph does show that the bars representing the compressive strength of the 10% SSA mixture on day 7 and day 28 are roughly the same height (suggesting that the mixture's compressive strength changed by only a small amount), and although there are relatively large gaps between the bars showing the compressive strength of the 20% and 40% SSA mixtures on day 7 and day 28 (suggesting that their compressive strength changed over time by a greater amount), this comparison concerns only how each SSA mixture's strength changed over time and doesn't compare the SSA mixtures with the control mixture. Supporting the recommendation that SSA substitution be done in small

proportions requires evidence that small substitutions don't compromise strength relative to the control while larger substitutions do. Choice C is incorrect because it doesn't accurately represent the data in the graph, nor is the comparison it makes relevant to the researchers' recommendation. While the graph does show that the compressive strength of the 10% SSA mixture was greater than that of the control on day 7 (about 25 megapascals vs. about 22 megapascals) and, moreover, that the compressive strength of the control increased from day 7 to day 28 (from about 22 megapascals to about 26 megapascals), the graph shows that the compressive strength of the 10% SSA mixture increased by a small amount (from about 25 megapascals to slightly more than 25 megapascals), not that it decreased. Moreover, the researchers recommended substituting relatively small proportions (compared with larger proportions) of the mixtures with SSA; comparing the compressive strength of the 10% SSA mixture with that of the control without also considering how the compressive strength of the 10% SSA mixture compared with that of the 20% and 40% SSA mixtures wouldn't support the recommendation. Choice D is incorrect. Though the graph indicates that the 10% SSA mixture's compressive strength on day 7 and day 28 (about 25 megapascals and slightly more than 25 megapascals, respectively) didn't surpass the control mixture's compressive strength on day 28 (about 26 megapascals), this observation doesn't support the recommendation. The recommendation concerns the trade-off between SSA proportion and compressive strength; supporting it requires not just evidence about how the 10% SSA mixture compared with the control but also evidence about how the mixtures with larger proportions of SSA (20% and 40%) compared with the 10% SSA mixture and the control.

▼ 3.84 — B

Choice B is the best answer because it presents the quotation that best supports the student's claim about Braschi. By describing how Braschi's blending of genres invites her audience to think about how other art forms could also engage with literature, the quotation supports the idea that the diversity of responses to Braschi's work reflects Braschi's own approach to creating literature.

Choice A is incorrect because the quotation describes scholars from different countries writing essays about Braschi's use of language in her writings; it doesn't address how Braschi's creation of cross-genre literature inspires diverse types of responses, which is the claim the student makes. Choice C is incorrect because the quotation focuses on the fact that Braschi studied in several different cities, which doesn't address the student's claim that Braschi's creation of cross-genre literature inspires diverse types of responses. Choice D is incorrect because the quotation lists some of the authors who Braschi has written

academic works about, which is irrelevant to the student's claim that Braschi's creation of cross-genre literature inspires diverse types of responses.

▼ 3.85 — B

Choice B is the best answer. The claim is that the lava freed the deep nitrate, allowing it to float upward. The table supports this by showing that there was more nitrate in the lava-affected seawater at various depths above 300 meters (the depth to which the lava plunged) than in unaffected seawater.

Choice A is incorrect. This choice doesn't complete the statement in a way that supports the claim. The claim is about nitrate, not phosphate. Choice C is incorrect. This choice doesn't complete the statement in a way that supports the claim. The claim is only about nitrate—the concentrations of phosphate aren't relevant. Choice D is incorrect. This choice doesn't complete the statement in a way that supports the claim. It doesn't say anything about the seawater inside the lava-affected area.

▼ 3.86 — C

Choice C is the best answer. The graph shows that the CNO-treated females were way more active than the CNO-treated males, while the saline-treated males and females (the control groups) had very similar activity levels. This supports the claim that there were sex-specific differences in the mice's response to neuron activation.

Choice A is incorrect. This choice misreads the graph. All four groups of mice started at nearly the same activity level before treatment (see how all four points are very close together at -4 minutes, meaning four minutes before treatment). Choice B is incorrect. This choice doesn't complete the assertion. The assertion is about the mice's response to neuron activation, so we need to include the data about the CNO-treated females and males. Choice D is incorrect. This choice misreads the graph. The line for the CNO-treated males does not show a "substantial decline" until around 122 minutes, and there is no "steep increase" afterward.

▼ 3.87 — A

Choice A is the best answer because it presents the quotation that best supports the student's claim that in *The Politics*, Aristotle gives advice on preserving constitutions—preventing governments from falling and maintaining order—and specifically asserts that in a healthy state, laws must be followed as strictly as possible and infractions should not be overlooked even if they are minor. The philosopher states that when Aristotle builds his

argument about the characteristics of a well-functioning government, Aristotle asserts that transgression, or violation of law, will ruin the state if it "creeps in unperceived," or goes unnoticed. The philosopher then adds that Aristotle illustrates this point by comparing the situation to one in which small but frequent expenses diminish a fortune almost unnoticeably until, eventually, the fortune is entirely gone. In other words, the philosopher indicates that Aristotle makes the point that total obedience to law preserves a healthy state while even small violations, if ignored, will undermine the health of the state.

Choice B is incorrect because the philosopher addresses Aristotle's observation about corruption within the government (in particular, preventing the possibility that members of the government can take bribes), and although corruption can involve infractions, the observation is about a subset of people within the state and isn't directly connected to the importance of upholding total obedience to the law throughout the state. Choice C is incorrect because the philosopher discusses Aristotle's point about those who would intentionally destroy a constitution altogether and the need for rulers to remind the populace that it would be dangerous for a constitution to collapse, but neither idea is directly connected to the importance of upholding total obedience to the law. Choice D is incorrect because the philosopher explains that Aristotle makes the point that oligarchic leaders may retain power by having members of disenfranchised classes participate in government alongside governing classes, and this point doesn't address the importance of ensuring obedience to the law and addressing even minor violations.

▼ 3.88 — B

Choice B is the best answer because it presents a finding that, if true, would most directly support the researchers' hypothesis that the smaller maximum size of PCFG whales is an adaptation to resource opportunities particular to their foraging range. The text explains that PCFG whales, a subgroup of ENP gray whales, forage along the coasts of Northern California and British Columbia rather than in the Arctic, where the main ENP group forages. The text goes on to state that PCFG whales reach smaller maximum sizes than other ENP gray whales despite having similar pre-maximum growth rates. If PCFG whales tend to forage in shallow, rocky kelp beds that whales as large as those in the main group cannot access, that would indicate that smaller body size confers a direct advantage in this foraging range—namely, access to a food source that larger whales cannot reach.

Choice A is incorrect because a finding describing recent trends in average body size for both groups of whales wouldn't address whether the smaller maximum size of PCFG whales reflects an adaptation to their foraging environment. Recent stability or change in average

size doesn't speak to why the size difference between PCFG whales and the main ENP group exists. Choice C is incorrect because a finding about PCFG whales' proximity to ports and urban populations wouldn't address the resources available to PCFG whales or the relationship between body size and foraging. Proximity to human activity is irrelevant to a hypothesis about body size being an adaptation to resource opportunities. Choice D is incorrect. Although a finding about a certain prey species being available in the PCFG whales' foraging waters but not in the main group of ENP whales' foraging waters would establish that the PCFG foraging range offers distinct prey species, it wouldn't connect the availability of those prey species to body size in a way that would explain the PCFG whales' smaller size. Without indicating that the available prey in the PCFG range favors smaller whales, the finding falls short of supporting the hypothesis that smaller size is an adaptation to resource opportunities.

▼ 3.89 — A

Choice A is the best answer because it presents a finding that, if true, would most strongly support the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were. The text explains that Vaiglova, Liu, and their colleagues analyzed the chemical composition of sheep, goat, and cattle bones from the Bronze Age in China in order to investigate the animals' domestication, or their adaptation from a wild state to a state in which they existed in close connection with humans. According to the text, the team's analysis showed that sheep and goats of the era fed largely on wild plants, whereas cattle fed on millet—importantly, a crop cultivated by humans. If analysis of the animal bones shows that the cattle's diet also consisted of wheat, another crop cultivated by humans in China during the Bronze Age, the finding would support the team's conclusion by offering additional evidence that cattle during this era fed on human-grown crops—and, by extension, that humans raised cattle relatively close to the settlements where they grew these crops, leaving goats and sheep to roam farther away in areas with wild vegetation, uncultivated by humans.

Choice B is incorrect because if it were true that sheep's and goats' diets consisted of small portions of millet, which the text states was a crop cultivated by humans, the finding would suggest that sheep and goats were raised relatively close to human settlements, weakening the team's conclusion that cattle were likely raised closer to those settlements than sheep and goats were. Choice C is incorrect because the finding that cattle generally require more food and nutrients than do sheep and goats wouldn't support the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were. Nothing in the text suggests that cattle were incapable of obtaining sufficient food and nutrients without

access to human-grown crops. Hence, even if cattle's diets are found to have different requirements than the diets of sheep and goats, the cattle could have met those requirements from food located far from human settlements. Choice D is incorrect because if it were true that the diets of sheep, goats, and cattle varied based on what the farmers in each Bronze Age settlement could grow, the finding would weaken the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were, suggesting instead that all three types of animals were raised close enough to human settlements to feed on those settlements' crops.

▼ 3.90 — D

Choice D is the best answer because it provides a quotation that effectively supports the student's claim about the film *One Night in Miami…*. The quotation states that in researching the play on which the film was based, Kemp Powers only found superficial details about what actually happened during the 1964 meeting in Miami between four leading Civil Rights leaders, meaning that there is very little information about the meeting in the historical record. In the absence of greater details, it wouldn't have been possible for the film to be a precise retelling of the historical events it depicts. The quotation explains that to compensate for this lack of information about the meeting, Powers did extensive research into the four figures and how they thought at the time in order to speculate in an informed way about what they might have said or what might have occurred between them. Therefore, the quotation effectively supports the claim that the film is best understood not as a precise retelling of a historical event but as a deeply informed imaginative rendering of that event.

Choice A is incorrect. Although the quotation discusses how on learning about the 1964 meeting in Miami, Powers was inspired to write a play and, later, to adapt it into a screenplay, it doesn't discuss Powers's approach to representing what had occurred in the meeting. Instead, it states that Powers didn't initially plan to write a story only "focusing on the meeting itself" but rather had considered writing a "much longer" and more expansive work about the meeting's four participants. Choice B is incorrect because the quotation doesn't discuss Powers's approach to representing historical events in his play and in the film; instead, the quotation focuses on the film's positive critical reception by mentioning that it received numerous awards and nominations. Choice C is incorrect. Although the quotation references historical events that are discussed directly in the play and film by explaining how the four historical figures featured in the story engage in political debates about contemporary issues, it doesn't specify to what extent Powers's representation of what occurred during the 1964 meeting in Miami is a factual retelling of what happened and how much is an imaginative rendering of what might have happened. Rather, the quotation

focuses on Powers's description of the film's basic premise and how the characters engage with the historical context of its setting.

▼ 3.91 — A

Choice A is the best answer because it most effectively uses data from the graph to complete the student's conclusion about beehive structure. The text explains that in the hives of honeybees, the hexagonal cells housing drone eggs are larger than the hexagonal cells housing worker eggs, and that this size difference results in a construction problem that the bees address by using nonhexagonal cells to fill gaps between sections of drone-egg cells and worker-egg cells. The text also states that the size difference between drone-egg cells and worker-egg cells varies by species of honeybee. The graph displays data on the percentage of nonhexagonal cells in the hives of three species. In the hives of the western honeybee, the percentages of five-sided, seven-sided, and eight-sided cells are all less than 0.5%. But in the hives of the black dwarf honeybee, the percentages of five-sided and seven-sided cells are higher than those for the western honeybee: about 2.5% for both. And for the dwarf honeybee, the percentages of five-sided and seven-sided cells are also higher than those for the western honeybee: slightly over 2.5% and slightly over 2.0%, respectively; additionally, the dwarf honeybee possesses a higher percentage of eight-sided cells than the western honeybee does. Taken altogether, the graph shows that the hives of the western honeybee consist of a smaller percentage of nonhexagonal cells than the hives of the two other species do. Since the nonhexagonal cells exist only to solve the construction problem arising from the difference in size between drone-egg cells and worker-egg cells, a smaller percentage of nonhexagonal cells would be associated with a smaller size difference between the two types of cells. Therefore, it can be concluded from the data that worker-egg cells are probably closer in size to drone-egg cells in the hives of the western honeybee than in the hives of the other two species.

Choice B is incorrect because, as the text states, honeybee species deposit their eggs in hexagonal cells, not in nonhexagonal ones. Thus, the western honeybee and black dwarf honeybee wouldn't deposit drone eggs in eight-sided cells, and the dwarf honeybee wouldn't deposit drone eggs in seven-sided cells. Choice C is incorrect. The text explains that honeybees rely mainly on one geometric shape, the hexagon, when constructing their hives, and the graph shows that the western honeybee relies on the same nonhexagonal shapes as the dwarf honeybee does: five-sided, seven-sided, and eight-sided cells. In other words, the western honeybee and dwarf honeybee rely on the same number of geometric shapes. For the black dwarf honeybee, the graph displays data only for five-sided and seven-sided cells, which suggests a total absence of eight-sided cells. Yet this would be only one less

nonhexagonal shape than is seen in the western honeybee. Thus, based on the graph, it would be inaccurate to say that the western honeybee relies on "many more" geometrical shapes than the other two species do. Choice D is incorrect. As the text explains, honeybee hives consist mainly of hexagonal cells, and sections of nonhexagonal cells are used to connect sections of hexagonal cells of different sizes. Since the graph indicates that the percentage of nonhexagonal cells is lower for the western honeybee than it is for the dwarf honeybee or black dwarf honeybee, the western honeybee would conversely have a *higher* percentage of hexagonal cells than the either the dwarf honeybee or black dwarf honeybee does, not a lower percentage.